RANDOM WALK ON SPHERE PACKINGS AND DELAUNAY TRIANGULATIONS IN ARBITRARY DIMENSION

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ABSTRACT. We prove that random walks on a family of tilings of d-dimensional Euclidean space, with a canonical choice of conductances, converge to Brownian motion modulo time parameterization. This class of tilings includes Delaunay triangulations (the dual of Voronoi tesselations) and sphere packings. Our regularity assumptions are deterministic and mild. For example, our results apply to Delaunay triangulations with vertices sampled from a d-dimensional Gaussian multiplicative chaos measure. As part of our proof, we establish the uniform convergence of certain finite volume schemes for the Laplace equation, with quantitative bounds on the rate of convergence. In the special case of two dimensions, we give a new, short proof of the main result of Gurel-Gurevich, Jerison, and Nachmias [2020].

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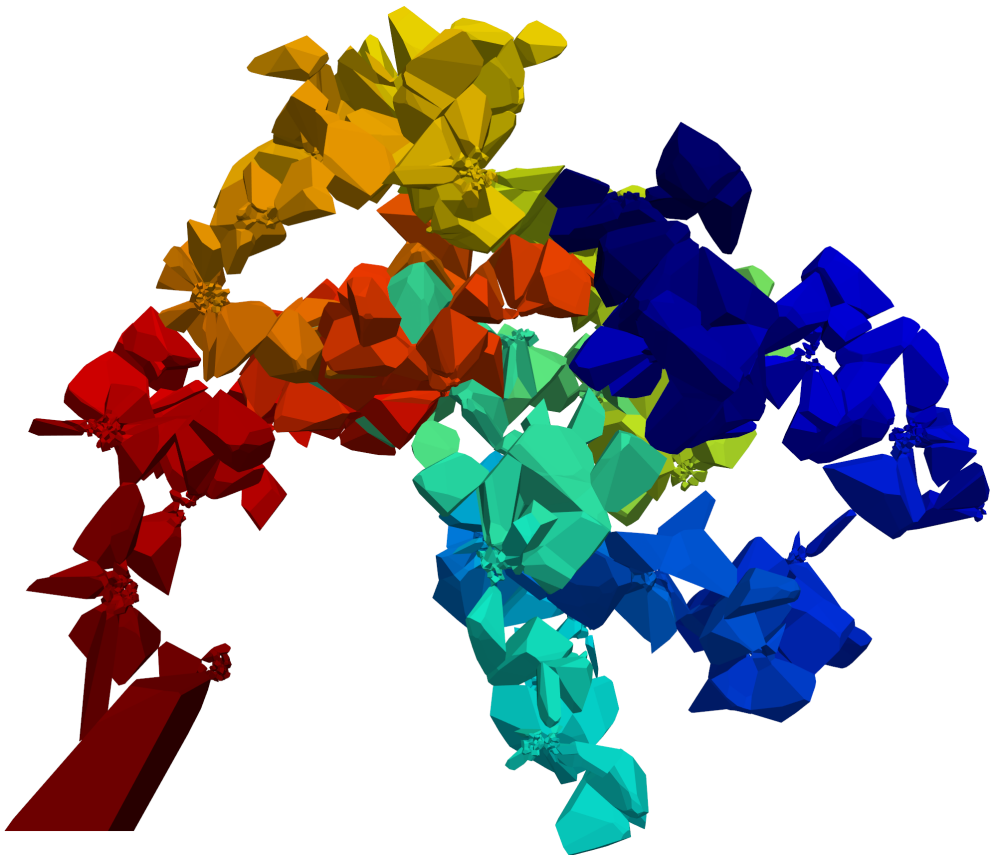


FIGURE 1. The trace of random walk on the Voronoi tessalation of 2^{19} points sampled from a Gaussian multiplicative chaos measure on a cube with parameter $\gamma = 1$. The random walk is stopped upon exiting the cube and the cells of the tessalation are colored according to the order in which they are visited.

1. INTRODUCTION

In this paper, we consider a class of graphs embedded in $\mathbb{R}^d$, called *orthogonal tilings*, which come equipped with a canonical choice of conductances on their edges (defined in Section 1.1). This class includes tangency graphs of sphere packings as well as d-dimensional Delaunay triangulations (the dual graphs of Voronoi tesselations). We show that under nearly minimal regularity assumptions, random walks on these graphs converge to Brownian motion modulo time change, equivalently, discrete harmonic functions converge to their continuum counterparts (Theorems A and B). Several motivations drive this work, and these are discussed in detail in Sections 1.3 and 1.4; to summarize:

- Orthogonal tilings in $d = 2$ have been studied in part to establish universality of statistical mechanics models, see, for example, the ICM survey, Smirnov [2010]. Our work may be viewed as the first step in this program in $d \geq 3$.
- We obtain a higher-dimensional analog of the results of Gurel-Gurevich et al. [2020], which gives the convergence of discrete harmonic functions on planar orthodiagonal lattices to their continuum counterparts (see also Dubejko [1999], Chelkak and

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Smirnov [2011], Skopenkov [2013], and Werness [2015] for earlier results). In the special case when $d = 2$, we also obtain a short new proof of the main result of Gurel-Gurevich et al. [2020].

- Theorem B can be viewed as a uniform convergence statement for a class of finite volume approximation schemes for the Dirichlet problem. Such approximation schemes are widely used in numerical analysis, see, for example the textbook Eymard, Gallouët, and Herbin [2000]. For the class of approximation schemes considered in this paper, convergence was previously only known in L^2, which is not sufficient to get convergence of the associated random walk.
- Theorems A and B apply to highly inhomogeneous tilings, including certain discretizations of higher-dimensional analogs of Liouville quantum gravity (LQG). The statement that random walk on such discretizations converges to Brownian motion modulo time change can be viewed as a higher-dimensional analog of scaling limit results for random walk on discretizations of LQG in two dimensions Gwynne, Miller, and Sheffield [2021, 2022], although the setup and proofs in the present paper are quite different.
- Our results provide some evidence that sphere packings and Delaunay triangulations are discrete analogs of conformally flat Riemannian metrics, which suggests that such graphs could possibly play a role analogous to planar maps in higher-dimensional random conformal geometry.

1.1. **Main results.** Let $\mathcal{D}$ be a subset of $\mathbb{R}^d$, cover $\mathcal{D}$ by closed, convex polytopes $\{P_v\}$ with disjoint interiors, and associate each polytope P_v with a vertex v in its interior. Denote the set of such vertices by $\mathcal{V}$ and declare that $v, w \in \mathcal{V}$ are joined by an edge if $P_v \cap P_w$ is a $(d-1)$-dimensional facet. Assume that each compact subset of $\mathcal{D}$ intersects finitely many polytopes. We call the resulting graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ a *tiling graph* of $\mathcal{D}$. An *orthogonal* tiling graph satisfies the additional property that for each edge $(w, v) \in \mathcal{E}$ the vector $(w - v)$ is orthogonal to the hyperplane containing $P_w \cap P_v$. As we observe in Section 1.2 below, examples of orthogonal tilings include Voronoi tessellations and tangency graphs of sphere packings.

In this paper we investigate the large scale behavior of random walk on orthogonal tilings. The random walk $\{X_t\}_{t \geq 0}$ we consider is discrete time and time homogeneous with transition probabilities p determined by the geometry of the graph: for an edge $(w, v) \in \mathcal{E}$, we set

$$p(w,v) := \frac{\mathbf{a}(w,v)}{\mathbf{a}(w)}, \quad \text{where } \mathbf{a}(w,v) := \frac{\mathbf{vol}_{d-1}(P_w \cap P_v)}{\|w - v\|} \quad \text{and} \quad \mathbf{a}(w) := \sum_{(w,v) \in \mathcal{E}} \mathbf{a}(w,v). \tag{1.1}$$

Here, $\mathbf{vol}_{d-1}$ denotes $d-1$-dimensional Lebesgue measure. See Figure 2.

These probabilities "tilt" the random walk so that its Euclidean position is a discrete-time martingale, see Lemma 2.2 and [Gurel-Gurevich and Seidel, 2022, Proposition 6.4]. In fact, simple random walks on orthogonal tilings do not necessarily converge to Brownian motion modulo time change, even under very strong regularity assumptions — see Theorem 6.1 below.

The main result of this paper gives the convergence of $\{X_t\}_{t \geq 0}$ to Brownian motion modulo time parametrization. We recall the definition of the topology on curves modulo time parametrization from [Aizenman and Burchard, 1999, Section 2]. Let $T_1, T_2 > 0$ and

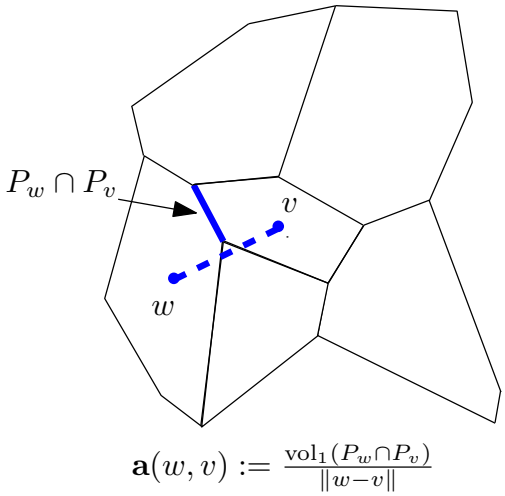


FIGURE 2. A polytope P_v and some of its neighbors in an orthogonal tiling. The edge between v and w (blue dots) is a blue dotted line and the facet $P_w \cap P_v$ is a thick blue line. The conductance $\mathbf{a}(w,v)$ is the length of the facet divided by the Euclidean distance $\|w-v\|$.

let $\phi_1 : [0, T_1] \to \mathbb{R}^d$ and $\phi_2 : [0, T_2] \to \mathbb{R}^d$ be two continuous curves. We define

$$d(\phi_1, \phi_2) := \inf_{\psi} \sup_{t \in [0,T_1]} |\phi_1(t) - \phi_2(\psi(t))|, \tag{1.2}$$

where the infimum is taken over all increasing homeomorphisms $\psi : [0, T_1] \to [0, T_2]$.

Theorem A (Convergence of random walk)**.** *Let $U \subset \mathbb{R}^d$ be a bounded, Lipschitz* [1] *domain such that $\overline{U} \subset \mathcal{D}$ and let $\{\mathcal{G}_n\} = \{(\mathcal{V}_n, \mathcal{E}_n)\}$ be a sequence of orthogonal tilings of $\mathcal{D}$. For $z \in \mathcal{D}$, let z^n be the nearest vertex in $\mathcal{G}_n$ to z with ties broken in lexicographical ordering. Assume that*

$$\lim_{n\to\infty} \left(\varepsilon_n + \sup_{z \in \mathcal{D}} \|z^n - z\| \right) = 0, \quad \text{where} \quad \varepsilon_n := \sup_{v \in \mathcal{V}_n} \mathrm{diam}(P_v) \tag{1.3}$$

and at least one of the following three conditions is satisfied:

- *(I) planarity, $d = 2$; or*
- *(II) the volume of the smallest tile in $\mathcal{G}_n$ is at least $exp(-o(\varepsilon_n^{-1}))$; or*
- *(III) there exists $\alpha > 0$ such that the diameter of each tile P_v in $\mathcal{G}_n$ is at most $O(\max_{(v,w)\in\mathcal{E}_n} \|w - v\|^\alpha \, \mathbf{vol}_{d-1}(P_w \cap P_v)^\alpha)$, where the max is over all edges incident to v.*

Then, for each $z \in U$, as $n \to \infty$, the linearly interpolated random walk $\{X_t\}$ started at z^n and stopped upon exiting U converges in law to standard Brownian motion started at z and stopped upon exiting U with respect to the metric on curves viewed modulo time parametrization (1.2). Moreover, the convergence is uniform over all choices of z.

[1] We need to assume the domain is Lipschitz because we need that Brownian motion started near the boundary exits the domain close to its starting point. Without this condition, the random walk could exit the domain much earlier than the Brownian motion.

The hypothesis (1.3) is close to necessary for the conclusion of Theorem A to hold. The hypotheses (II) and (III) are quite mild and are true for essentially any models one might be interested in, even discretizations of rough, fractal geometries (see Proposition 1.1). However, we do not know whether it is necessary to have one of these two hypotheses for $d \geq 3$ (see Problem 1.7).

Hypothesis (II) can be replaced by a slightly weaker hypothesis concerning the diameters of the connected components of the set of small tiles of $\mathcal{G}_n$, see Theorem 3.6. This weakened hypothesis is similar to [Chelkak, Laslier, and Russkikh, 2023, Assumption 1.2]. We emphasize that in Hypothesis (III), we allow any $\alpha > 0$. For $\alpha \neq 1/d$, the ratio $\|w-v\|^{\alpha}\,\mathbf{vol}_{d-1}(P_w \cap P_v)^{\alpha}/\mathrm{diam}(P_v)$ is not scale invariant, so the bound we require is much weaker than a uniform control over the geometry of the tiling at all scales.

The macroscopic features of a discrete time Markov process are closely tied to the large scale behavior of its generator, see, *e.g.*, [Kallenberg, 2021, Theorem 19.25]. The generator of X_t is

$$\Delta_{\mathbf{a}}^{\mathcal{G}} h(v) = \sum_{(w,v)\in\mathcal{E}} \mathbf{a}(w,v)(h(w)-h(v)) \quad \text{where} \quad \mathbf{a}(w,v) := \frac{\mathbf{vol}_{d-1}(P_w \cap P_v)}{\|w-v\|}, \tag{1.4}$$

as in (1.1). Theorem A is roughly equivalent to the statement that $\Delta_{\mathbf{a}}^{\mathcal{G}}$-harmonic functions are approximated uniformly, at large scales, by continuum harmonic functions.

In what follows, we write $\mathcal{V}_n[U]$ and $\partial\mathcal{V}_n[U]$ for the set of vertices of $\mathcal{G}_n$ in the interior of U and those which share an edge with a vertex in the interior, respectively.

Theorem B (Convergence of the Dirichlet problem)**.** *Let $U \subset \mathcal{D}$ be a bounded domain and let h_C be a function which is harmonic on U. Assume that at least one of the following two conditions hold:*

(a) h_C is harmonic in a neighborhood of $\overline{U}$; or
(b) h_C is continuous on $\overline{U}$ and U is Lipschitz and $\overline{U} \subset \mathcal{D}$.

If $\mathcal{G}_n$ satisfies (1.3) and one of (I), (II), or (III) in Theorem A, then the discrete harmonic extension h_D^n of h_C to $\mathcal{V}_n[U]$, defined by, [2]

$$\begin{cases} \Delta_{\mathbf{a}}^{\mathcal{G}} h_D^n = 0 & \text{in } \mathcal{V}_n[U] \\ h_D^n = h_C & \text{on } \partial\mathcal{V}_n[U]\,, \end{cases}$$

converges uniformly to h_C,

$$\lim_{n\to\infty} \sup_{z\in\mathcal{V}_n[U]} |h_D^n(z) - h_C(z)| = 0\,. \tag{1.5}$$

In fact, we obtain a quantitative bound on the rate of convergence under Assumption (a) in (1.5), see Theorems 3.3, 3.6, and 3.7 below.

1.2. **Examples.**

[2] In the case h_C is not defined at a boundary vertex v, pick the nearest point in the cell P_v for which it is defined.

1.2.1. *Sphere packings.* A *sphere packing* is a collection of d-dimensional spheres in $\mathbb{R}^d$ with disjoint interiors. The spheres are not required to have the same radii. A *sphere-packed graph*, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, is the tangency graph of a sphere packing. Each vertex $v \in \mathcal{V}$ is associated to a sphere S_v with v as its centerpoint, and for each edge $(v, w) \in \mathcal{E}$ there is a unique hyperplane $R_{v,w}$ which is tangent to both S_v and S_w at the unique point of $S_v \cap S_w$. A vertex v is *covered* if the intersection of the hyperplanes $\{R_{v,w}\}_{(v,w)\in\mathcal{E}}$ is a bounded d-dimensional polytope P_v containing v. A sphere-packed graph is *covering* if every vertex is covered. A sphere packing *covers* a domain U if the spheres in the packing cover U and every vertex whose corresponding sphere intersects U is covered. It is immediate from the definitions that a locally finite sphere packed graph which covers a domain is an orthogonal tiling of that domain.

1.2.2. *Voronoi tesselations and Delaunay triangulations.* The *Voronoi tesselation* of a locally finite set of points $S \subset \mathbb{R}^d$, is the partition of $\mathbb{R}^d$ into cells $\{\mathcal{C}_s\}_{s\in S}$ such that every point in the tile $\mathcal{C}_s$ is closer to s than any other point $s' \in S$,

$$\mathcal{C}_s = \{y \in \mathbb{R}^d : \|s - y\| \le \|s' - y\| \quad \forall s' \in S\}\,.$$

Equivalently, see, *e.g.*,[Møller, 1994, Section 1], the cell $\mathcal{C}_s$ is the intersection, over all $s' \in S$ with $s' \neq s$ of the closed halfspace containing s and bounded by the bisecting hyperplane of s and s',

$$\mathcal{C}_s = \bigcap_{s'\in S: s'\neq s} \{y \in \mathbb{R}^d : (y - \frac{1}{2}(s + s')) \cdot (s - s') \ge 0\}\,. \tag{1.6}$$

Consequently, cells are closed d-dimensional convex sets; and for any two neighbors s, s', the vector $(s - s')$ is orthogonal to the $(d-1)$-dimensional facet $\mathcal{C}_s \cap \mathcal{C}_{s'}$. If the cells are finite we denote the orthogonal tiling induced by the Voronoi tesselation of a point set $S \subset \mathbb{R}^d$ by

$$\mathcal{G}^{\mathrm{Vor}}(S) = (\mathcal{V}^{\mathrm{Vor}}(S), \mathcal{E}^{\mathrm{Vor}}(S))\,. \tag{1.7}$$

This is the *Delaunay triangulation* of the Voronoi tesselation.

Let $U \subset \mathbb{R}^d$ be a bounded domain. Let μ be a Radon measure defined in a neighborhood of $\overline{U}$ which satisfies, for some exponents $\beta^{\pm} > 0$, for all sufficiently small $r \in (0, 1)$, the bound

$$r^{\beta^+} \le \mu(B_r(z)) \le r^{\beta^-} \qquad \forall z \in U, \tag{1.8}$$

where $B_r(s)$ is the Euclidean ball of radius r centered at $s \in \mathbb{R}^d$. The following proposition implies that Hypothesis (II) is satisfied by $\mathcal{G}^{\mathrm{Vor}}(\Lambda_m)$, where Λ_m is a Poisson point process of intensity $m\mu$, for large m almost surely.

Proposition 1.1. *Let U be a bounded domain. With probability one, for all sufficiently large m, the cells of the Voronoi tesselation of Λ_m which intersect U satisfy,*

$$B_{m^{-8/\beta^-}}(s) \subset \mathcal{C}_s \subset B_{m^{-1/(3\beta^+)}}(s) \qquad \forall s \in \Lambda_m \text{ with } \mathcal{C}_s \cap U \neq \emptyset\,. \tag{1.9}$$

In particular,the conclusions of Theorem B and, if U is Lipschitz, Theorem A, hold for the graphs $\mathcal{G}^{\mathrm{Vor}}(\Lambda_m)$ as $m \to \infty$.

Remark 1.2. *Let Φ be a log-correlated Gaussian field on a bounded domain $U \subset \mathbb{R}^d$ and let $\gamma \in (0, \sqrt{2d})$. Let $\mu =$ "$e^{\gamma\Phi(x)}\,dx_1 \dots dx_d$" be the associated Gaussian multiplicative chaos (GMC) measure. It follows from standard estimates for GMC that there exists $\beta^{\pm} =$*

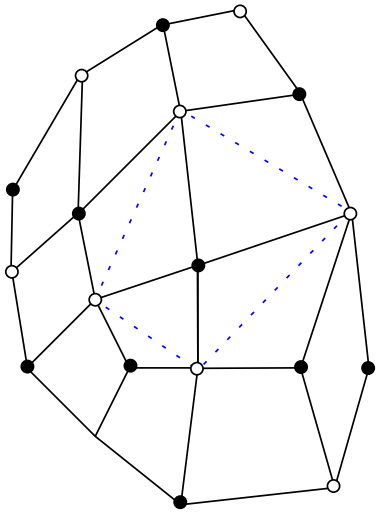


FIGURE 3. An orthodiagonal map. The vertices in $\mathcal{V}^\bullet$ ($\mathcal{V}^\circ$) are black (white) and the edges $\mathcal{E}$ are black lines. The polytope corresponding to the center vertex is outlined in dotted blue.

$\beta^\pm(d,\gamma) > 0$ such that a.s. (1.8) holds for μ for all sufficiently small $r \in (0,1)$ (how small is random). For example, this can be deduced from [Berestycki and Powell, 2024, Theorem 3.26] and a union bound argument. Hence, Proposition 1.1 applies to GMC measures on domains in $\mathbb{R}^d$.

1.2.3. *Orthodiagonal maps.* An *orthodiagonal* map is a finite connected planar graph in which each edge is a straight line segment, each inner face is a quadrilateral with orthogonal diagonals; and the boundary of the outer face is a simple closed curve. An orthodiagonal map can be expressed as $\mathcal{G} = ([\mathcal{V}^\bullet, \mathcal{V}^\circ], \mathcal{E})$, where $\mathcal{V}^\bullet, \mathcal{V}^\circ$ is a bipartition of the vertices of the graph. The vertices $\mathcal{V}^\bullet$ induce a graph $\mathcal{G}^\bullet = (\mathcal{V}^\bullet, \mathcal{E}^\bullet)$ where there is an edge between two vertices $v_1, v_2 \in \mathcal{V}^\bullet$, if they lie on the same inner face of $\mathcal{G}$. The vertices $\mathcal{V}^\circ$ similarly induce a graph $\mathcal{G}^\circ = (\mathcal{V}^\circ, \mathcal{E}^\circ)$. Isoradial graphs, discussed at the beginning of Section 1.4 below, can be represented as an orthodiagonal map, see [Gurel-Gurevich et al., 2020, Section 2.2].

The conductance of an edge (v_1, v_2) corresponding to the inner face of a quadrilateral (v_1, w_1, v_2, w_2) is given by

$$\mathbf{a}(v_1, v_2) := \frac{\|w_1 - w_2\|}{\|v_1 - v_2\|}\,.$$

It was shown in Gurel-Gurevich et al. [2020] that functions $f : \mathcal{V}^\bullet \to \mathbb{R}$ which are $\mathbf{a}$-discrete harmonic, that is, which satisfy,

$$\sum_{(v,w)\in\mathcal{E}^\bullet} \mathbf{a}(v,w)(f(v) - f(w)) = 0\,,$$

are close to continuous harmonic functions (in the sense of Theorem B).

When every quadrilateral in $\mathcal{G}$ is convex, $\mathcal{G}^\bullet$ can be represented as an orthogonal tiling. Each vertex $v \in \mathcal{V}^\bullet$ which is not on the outer face of $\mathcal{G}$ has a polytope P_v with edges given by the opposite diagonals of the quadrilaterals containing v — that is for each internal face (v, w, v', w') of $\mathcal{G}$ with $v' \in V^\bullet$ and $w, w' \in V^\circ$ the edge (w, w') — see Figure 3. It is

straightforward to see for each $v, v' \in \mathcal{V}^\bullet$ that $\mathbf{a}(v, v')$ coincides with the conductance defined in (1.4) and that convexity of each quadrilateral ensures v is contained in the interior of P_v and that P_v is a simple, convex polygon.

We describe how to adapt our proof to the case where non-convex quadrilaterals are allowed in Remark 3.4 below.

1.3. **Prior work and proof idea.** The equation (1.4) is a *finite volume scheme* on $\mathcal{G}_n$ and it is well known that the convergence in Theorem B holds (with no additional assumptions on $\mathcal{G}_n$) in L^2, see, *e.g.*, [Eymard et al., 2000, Theorem 9.3]. Briefly, a finite volume scheme is a numerical method for solving divergence-form elliptic partial differential equations. The domain is partitioned into "control volumes" and integrating the equation by parts on each volume leads to a finite set of constraints. In the case of the Laplacian on orthogonal tilings this gives the operator $\Delta_{\mathbf{a}}^{\mathcal{G}}$.

Our main contribution can be thought of as an L^2 to L^∞ estimate for the operator $\Delta_{\mathbf{a}}^{\mathcal{G}}$ under very weak regularity assumptions on the graph. As discussed above, these weak assumptions allow the result to apply to highly inhomogeneous random graphs arising from approximations of random fractal geometries.

L^2 to L^∞ estimates are established in previous work when $\Delta_{\mathbf{a}}^{\mathcal{G}}$ is *uniformly elliptic*, that is, when $\mathbf{a}$ is bounded uniformly from above and below, see for example [Delmotte, 1997, Proposition 5.3] for an interior L^2 to L^∞ estimate for discrete harmonic functions. The main difficulty in proving Theorem B is that, in general, the operator $\Delta_{\mathbf{a}}^{\mathcal{G}}$ is not uniformly elliptic *at any scale*.

We prove (1.5) by an iteration of an L^2 bound (Proposition 2.6) and a Poincaré type inequality (Lemma 3.1). The proof, although distinctly different, is reminiscent of the Campanato-type large-scale regularity iteration originating in Armstrong and Smart [2016] and exposited in [Armstrong, Kuusi, and Mourrat, 2019, Chapter 4]. The idea there, which has become a cornerstone of the theory of elliptic homogenization, is that if the solution of a heterogeneous equation can be approximated by a harmonic function across a wide range of scales, it should be as regular as harmonic functions.

Here our "harmonic approximation", Proposition 2.6, has an error that depends on the continuum harmonic function all the way up to the boundary. Since our graphs are possibly very irregular, there is no immediate way to "mollify away" this error. This prevents us from applying the iteration in Armstrong and Smart [2016].

Instead, we fix the continuum harmonic function and iterate the discrete approximation. We first show, by the L^2 estimate and a Poincaré-type inequality, that the discrete harmonic function, h_D, is pointwise close to the continuum one, h_C, on a dense set of columns (Proposition 3.2). The remaining "bad" set where $|h_D - h_C|$ is large has small measure. By Proposition 3.2 again, the discrete harmonic extension of h_C to the bad set is close to h_C on a set of even larger measure. Repeating this, using one of assumptions (II) or (III) in Theorem A to close the loop, yields the pointwise bound.

When $d = 2$, the case which has received the most attention in previous literature, closeness on a dense set of columns and the maximum principle immediately imply the pointwise bound everywhere, yielding a short proof of Theorem B. This two dimensional result slightly improves [Gurel-Gurevich et al., 2020, Theorem 1.1] in that we do not require the domains to be simply connected.

Earlier work in dimension two, for instance Dubejko [1999], Chelkak and Smirnov [2011], Skopenkov [2013], and Werness [2015], imposed various assumptions on the graph which ensure that the operator $\Delta_{\mathbf{a}}^{\mathcal{G}}$ is uniformly elliptic. A similar result for finite volume schemes of a convection diffusion equation in dimensions two and three was also proved (again under assumptions implying uniform ellipticity) in [Coudière, Gallouët, and Herbin, 2001, Corollary 1].

1.4. **Universality and random geometry.** A special case of orthogonal tilings in two dimensions are *isoradial graphs*, graphs which have an embedding into $\mathbb{R}^2$ so that every face F of the graph lies on a circle of radius 1 with center in the interior of F. The term was coined in Kenyon [2002] and a theory of discrete analytic functions on these graphs was developed in Duffin [1968] and later, independently in Mercat [2001].

Convergence of the Dirichlet problem on isoradial graphs, Chelkak and Smirnov [2011], was used in a companion paper, Chelkak and Smirnov [2012], to prove convergence of the critical Ising model, thus establishing *universality* for the model. In an ICM survey Smirnov [2010] asked if these results generalize to other two-dimensional graphs and higher dimensions; in particular, our work addresses [Smirnov, 2010, Questions 1 and 12]. This question was also highlighted again in [Skopenkov, 2013, Problem 5.9]. Several other models have been studied on isoradial graphs including critical dimers, loop-erased random walk, spanning trees, the random-cluster model, and bond percolation; see the survey Boutillier and de Tilière [2012] and the papers de Tilière [2007], Grimmett and Manolescu [2014], Boutillier, de Tilière, and Raschel [2017], Duminil-Copin, Li, and Manolescu [2018]. An interesting next step would be to consider these models for $d \geq 3$ on orthogonal tilings, see Problem 1.3 below.

This paper can also be seen as a contribution towards understanding higher-dimensional analogues of discrete conformal geometry and *Liouville quantum gravity* (LQG). Roughly speaking, LQG is a theory of random geometry in two dimensions which describes the scaling limit of discrete random surfaces, such as uniform triangulations of the sphere, see the surveys Gwynne [2020], Sheffield [2023] and the book Berestycki and Powell [2024].

There has been recent interest in establishing a corresponding theory in $d \geq 3$, see, for example, Schiavo, Herry, Kopfer, and Sturm [2021], B. Cerclé [2022], Ding, Gwynne, and Zhuang [2023]. These works study a notion of continuum random geometry in higher dimensions, which is described by a random Riemannian metric tensor of the form $g = e^{\gamma\Phi} g_0$, where $\gamma \in (0, \sqrt{2d})$ is a parameter, Φ is a log-correlated Gaussian field on a d-dimensional manifold (or a minor variant thereof), and g_0 is a fixed smooth background metric. If we take g_0 to be the flat metric, then (at least at a heuristic level) the random metric tensor g is *conformally flat*, meaning that it is a scalar function times the flat metric. It remains open to determine the correct discrete analog of this continuum theory (analogous to random planar maps in $d = 2$). See, *e.g.*, [Ding et al., 2023, Section 1.2]. The results of this paper contribute to this problem in two respects.

A natural approach to finding a discrete analog of LQG in d dimensions is to search for a class of graphs embedded in $\mathbb{R}^d$ which satisfy a discrete analog of "conformal flatness". In Benjamini and Curien [2011] sphere packings were studied partly to understand higher dimensional quantum gravity and it was asked, [Benjamini and Curien, 2011, Question 2], if being representable as the tangency graph of a sphere packing is a discrete analog of being "conformally flat". Theorem A may be viewed as positive evidence for this, since it implies that the scaling limit of random walk on sphere packings is Brownian motion modulo time

change, *i.e.*, in the scaling limit there is no directional bias depending on where the walk is in space.

Gaussian multiplicative chaos (GMC), Kahane [1985], Rhodes and Vargas [2014], Berestycki and Powell [2024], provides the analogue of the LQG area measure in $d \geq 3$. Voronoi tessellations with centers sampled according to a Poisson point process with respect to GMC (trivially) converge to GMC in the large sample size limit. Thus, Proposition 1.1 and Remark 1.2 give the convergence to Brownian motion (modulo time change) for random walk on certain embedded random graphs which approximate higher-dimensional analogs of LQG. The first papers to establish this sort of result in two dimensions were Gwynne et al. [2021, 2022] which prove the convergence of random walk on *mated-CRT maps*, a family of random planar maps known to converge to Liouville quantum gravity. The setting and proof of this result, however, are quite different from the present paper. The proof uses both planarity and the randomness of the environment (whereas here we have a fixed, deterministic environment in arbitrary dimension). Moreover, there are no conductances on the edges of the mated-CRT map, so the random walk is not a martingale (although the proofs show that it is approximately a martingale at large scales). See also Problem 1.4.

1.5. **Open questions.** We mention some possible directions for future research motivated by our work.

Problem 1.3. *What can be said about other statistical mechanics models, e.g., loop-erased random walk, uniform spanning tree, percolation, Ising model, dimers, discrete Gaussian free field, etc., on orthogonal tilings for $d \geq 3$?*

The paper Gwynne et al. [2022] proves that random walk converges to Brownian motion modulo time change in certain random planar environments which are not stationary with respect to translations, but are instead only "translation invariant modulo scaling". It is plausible that this result could be extended to dimension $d \geq 3$ if one assumes that the edge conductances scale by Λ^{d-2} when one scales space by Λ, see [Gwynne et al., 2022, Section 1.5]. It appears that the main obstacle to proving such an extension is transferring from an L^2 bound for discrete harmonic functions to an L^∞ bound (this corresponds to Lemma 2.19 in Gwynne et al. [2022], the proof of which strongly uses planarity).

Problem 1.4. *Can the techniques of this paper be used to prove a higher dimensional analogue of Gwynne et al. [2022]?*

A consequence of Theorem A is that random walk on Voronoi tessellations with centers sampled from a d-dimensional Gaussian multiplicative chaos measure converges to Brownian motion modulo time change. The paper Berestycki and Gwynne [2022] proves that random walk on mated-CRT maps converges in law with respect to the uniform topology to *Liouville Brownian motion*, the natural LQG time change of Brownian motion. It would be interesting to establish an analog of this result for the Voronoi tessellations considered in this paper.

Problem 1.5. *Determine the scaling limit of random walk on Voronoi tessellations with centers sampled from d-dimensional GMC viewed as a parametrized path which spends one unit of time in each cell.*

Instead of giving the walk its natural parametrization, we could alternatively seek a time parametrization which makes it converge to standard Brownian motion. To be consistent

with Brownian scaling, we need to parametrize so that if we scale space by Λ, the amount of time spent in each polytope scales like Λ^2.

Problem 1.6. *Consider random walk on an orthogonal tiling graph parameterized so that the amount of time it spends at each polytope is proportional to the volume of the polytope to the power $2/d$. Does this random walk converge to standard Brownian motion?*

Problem 1.7. *For $d \geq 3$, either show that the conclusion Theorem A is true without assumption (II) or (III); or give an explicit counterexample to show that at least one of these assumptions is necessary.*

1.6. **Paper outline.** We start in Section 2 with discrete PDE estimates, geometric bounds for orthogonal tilings, and the aforementioned L^2 estimate for $\Delta_{\mathbf{a}}^{\mathcal{G}}$-harmonic functions. All of the results contained in the section are standard but included for the reader's convenience. In Section 3 we prove Theorem B under Hypothesis (a) and at least one of Assumptions (I), (II), or (III). Then, in Section 4 we use this to prove Theorem A and subsequently deduce Theorem B under Hypothesis (b). In fact, there we prove a "black box" statement asserting, roughly, that whenever one has convergence of the Dirichlet problem, one has convergence of the associated random walk. In Section 5 we prove Proposition 1.1, which verifies that our results apply to graphs derived from Gaussian multiplicative chaos measures. Finally, in Section 6 we give an example of a sphere packing in $d \geq 3$ for which simple random walk (with unit conductance) does not converge to Brownian motion modulo time parameterization.

Conventions and notation.

- For a set $U \subset \mathbb{R}^d$, we denote the d-dimensional Lebesgue measure by $\mathbf{vol}_d(U)$ and we denote its Euclidean diameter by $\operatorname{diam}(U)$.
- For a point $v \in \mathbb{R}^d$, $\|v\|$ is the Euclidean norm.
- For a point $v \in \mathbb{R}^d$ and $r > 0$, we write $B_r(v)$ for the Euclidean ball of radius r centered at v. When $v = 0$, we write B_r.
- Unless explicitly mentioned, we henceforth fix a set $\mathcal{D}$ and an orthogonal tiling $\mathcal{G}$ of $\mathcal{D}$ and denote the set of edges and vertices by $\mathcal{E}$ and $\mathcal{V}$ respectively. We also denote, for $e = (w, v) \in \mathcal{E}$,
$$H_e := P_w \cap P_v \,.$$
- In an abuse of notation, we identify vertices $v \in \mathcal{V}$ with their location in $\mathbb{R}^d$ and the edge $(w, v) \in \mathcal{E}$ with the straight line between w and v. We also denote the length of the edge $e = (w, v) \in \mathcal{E}$, by $\|e\| := \|w - v\|$.
- The induced subgraph of a domain $U \subset \mathbb{R}^d$ is $\mathcal{G}[U]$. It is defined as the set of vertices contained in the interior of U together with the edges connecting pairs of vertices in that subset — see Figure 4. The edges and vertices of an induced subgraph are $\mathcal{E}[U]$ and $\mathcal{V}[U]$ respectively.
- The set of edges connecting pairs of vertices in a set A is denoted by $\mathcal{E}[A]$. The set of edges in $\mathcal{E}$ with at least one vertex in A is $\mathcal{E}[\overline{A}]$ and $\partial\mathcal{E}[A] = \mathcal{E}[\overline{A}] \setminus \mathcal{E}[A]$. In an abuse of notation, we consider $\mathcal{E}$ to be a set of both directed and undirected edges.
- The sets ∂U, U°, and $\overline{U}$ denote the boundary, interior, and closure of U, a subset of $\mathbb{R}^d$.
- For a collection of vertices $A \subset \mathbb{R}^d$, we write ∂A for the set of vertices in $\mathcal{V} \setminus A$ which share an edge with a vertex in A and $\overline{A} = A \cup \partial A$.

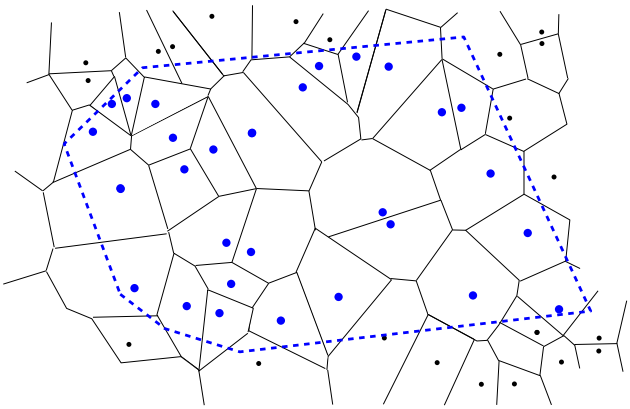


FIGURE 4. The domain U is outlined in dashed blue and the vertices of the induced subgraph $\mathcal{G}[U]$ are blue dots.

- For a domain $U \subset \mathbb{R}^d$ we define the set

$$\widehat{U} := \bigcup_{v \in \overline{\mathcal{V}[U]}} P_v \,, \tag{1.10}$$

  as the set of polytopes with vertices in or adjacent to U.
- A *vector field* θ is a function $\theta : \mathcal{E}[A] \to \mathbb{R}$ which is antisymmetric: $\theta(w,v) = -\theta(v,w)$ for every $(w,v) \in \mathcal{E}[A]$.

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2. Preliminaries

Here we collect, for completeness, some standard preliminary results that are essentially contained in, *e.g.*, the textbooks Eymard et al. [2000], Telcs [2006], Droniou, Eymard, Gallouët, Guichard, and Herbin [2018].

2.1. Discrete PDE.

We write $\boldsymbol{\Delta}$ and $\boldsymbol{\nabla}$ for the continuum Laplacian and gradient on $\mathbb{R}^d$. For a function $f : \mathcal{V} \to \mathbb{R}$ the vector fields $\nabla^{\mathcal{G}} f$ and $\nabla^{\mathcal{G}}_{\mathbf{a}} f$ are defined by, for a directed edge $e = (w,v)$,

$$\nabla^{\mathcal{G}} f(e) = f(w) - f(v) \tag{2.1}$$

and

$$\nabla^{\mathcal{G}}_{\mathbf{a}} f(e) = \mathbf{a}(e)(f(w) - f(v)) \,. \tag{2.2}$$

The *divergence* of a vector field $\theta : \mathcal{E} \to \mathbb{R}$ is the scalar function

$$(\mathbf{div}\, \theta)(v) := \sum_{(w,v) \in \mathcal{E}} \theta(e) \,, \tag{2.3}$$

where the sum is over all the directed edges whose endpoint is v. Observe that we can write

$$\Delta_{\mathbf{a}}^{\mathcal{G}} h(v) = (\mathbf{div}\, \nabla_{\mathbf{a}}^{\mathcal{G}} h)(v)\,.$$

We first recall a version of the discrete divergence theorem. This holds for general graphs and is well known, but we include a (short) proof.

Lemma 2.1. *Let $A \subset \mathcal{V}$ be a finite set of vertices, and let $f : \overline{A} \to \mathbb{R}$ be a function which is zero on ∂A. Then, for any vector field $\theta : \mathcal{E}[\overline{A}] \to \mathbb{R}$ we have that*

$$\sum_{v \in A} (\mathbf{div}\, \theta)(v) f(v) = -\frac{1}{2} \sum_{e \in \mathcal{E}[\overline{A}]} \theta(e) \nabla^{\mathcal{G}} f(e)\,. \tag{2.4}$$

Proof. Each neighboring pair of vertices $\{w, v\}$ appears twice in the sum on the right in (2.4), once as $\theta(w,v)(f(w) - f(v))$ and again as $\theta(v,w)(f(v) - f(w)) = \theta(w,v)(f(w) - f(v))$, by antisymmetry of θ. Consequently, since $f \equiv 0$ on ∂A,

$$\sum_{e \in \mathcal{E}[\overline{A}]} \theta(e) \nabla^{\mathcal{G}} f(e) = \sum_{v \in A} -2 f(v) \sum_{(w,v) \in \mathcal{E}} \theta(w,v) = \sum_{v \in A} -2 f(v) (\mathbf{div}\, \theta)(v)\,,$$

which completes the proof. □

The next lemma states that linear functions lie in the kernel of $\Delta_{\mathbf{a}}^{\mathcal{G}}$. In particular, like the usual Laplacian, the operator $\Delta_{\mathbf{a}}^{\mathcal{G}}$ has no drift. This is not used anywhere in the paper, but is included to give some justification for the choice of conductances in $\Delta_{\mathbf{a}}^{\mathcal{G}}$. A similar statement in the special case of sphere packings appears as [Gurel-Gurevich and Seidel, 2022, Proposition 6.4].

Lemma 2.2. *For every $p \in \mathbb{R}^d$ and $c \in \mathbb{R}$, denoting $\ell_p(x) = p \cdot x + c$, we have that $\Delta_{\mathbf{a}}^{\mathcal{G}} \ell_p = 0$.*

Proof. Let $v \in \mathcal{V}$ and observe that by definition,

$$\Delta_{\mathbf{a}}^{\mathcal{G}} \ell_p(v) = \sum_{(w,v) \in \mathcal{E}} \frac{\mathbf{vol}_{d-1}(P_w \cap P_v)}{\|w - v\|} (\ell_p(w) - \ell_p(v)) = \sum_{(w,v) \in \mathcal{E}} \mathbf{vol}_{d-1}(P_w \cap P_v) \left(p \cdot \frac{(w - v)}{\|w - v\|} \right)\,.$$

By the (continuum) divergence theorem we have

$$0 = \int_{P_v} \mathbf{\Delta} \ell_p = \int_{\partial P_v} \boldsymbol{\nabla} \ell_p \cdot \nu = \int_{\partial P_v} p \cdot \nu\,,$$

where ν denotes the outward pointing normal vector field to P_v. By the definition of an orthogonal tiling, whenever $(w,v) \in \mathcal{E}$ we have $\nu = \frac{w-v}{\|w-v\|}$ on $P_v \cap P_w$. Therefore,

$$\int_{\partial P_v} p \cdot \nu = \sum_{(w,v) \in \mathcal{E}} \mathbf{vol}_{d-1}(P_w \cap P_v) \left(p \cdot \frac{(w - v)}{\|w - v\|} \right)\,.$$

Combining the previous three displays completes the proof. □

We next recall the dual variational principle for the discrete Dirichlet problem. A continuum version of this appears in, for example, [Evans, 2010, Section 8.7, Exercise 13]. Also see [Gurel-Gurevich et al., 2020, Proposition 4.9]. This is only used in the proof of Proposition 2.6 below.

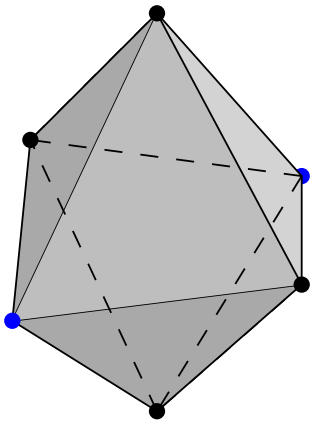


FIGURE 5. A dual polytope Q_e in three dimensions; the two vertices v and w are in blue and the extremal points of H_e are in black.

Lemma 2.3. *Let $A \subset \mathcal{V}$ be a finite set of vertices, and let $f : \overline{A} \to \mathbb{R}$ be a function which is zero on ∂A. Then, for any vector field $\theta : \mathcal{E}[\overline{A}] \to \mathbb{R}$ with $\mathbf{div}\,\theta = \mathbf{div}(\nabla^{\mathcal{G}}_{\mathbf{a}} f)$ on A we have*

$$\sum_{e \in \mathcal{E}[\overline{A}]} \mathbf{a}(e)^{-1} |\nabla^{\mathcal{G}}_{\mathbf{a}} f(e)|^2 \leq \sum_{e \in \mathcal{E}[\overline{A}]} \mathbf{a}(e)^{-1} \theta(e)^2 \,.$$

Proof. Define $\theta_* := \nabla^{\mathcal{G}}_{\mathbf{a}} f$ and let $\theta : \mathcal{E}[\overline{A}] \to \mathbb{R}$ be a vector field with $\mathbf{div}(\theta) = \mathbf{div}(\theta_*)$ on A and $\theta \neq \theta_*$. For $t \in [0,1]$ define

$$\phi(t) := \sum_{e \in \mathcal{E}[\overline{A}]} \mathbf{a}(e)^{-1} [\theta_*(e) + t(\theta(e) - \theta_*(e))]^2 \,,$$

and observe that it suffices to show that ϕ is strictly increasing on the interval $[0,1]$. We compute, for $t \in (0,1)$,

$$\begin{aligned}
\phi'(t) &= 2 \sum_{e \in \mathcal{E}[\overline{A}]} \mathbf{a}(e)^{-1} (\theta(e) - \theta_*(e)) [\theta_*(e) + t(\theta(e) - \theta_*(e))] \\
&= 2 \sum_{e \in \mathcal{E}[\overline{A}]} (\theta(e) - \theta_*(e)) \nabla^{\mathcal{G}} f(e) + 2t \sum_{e \in \mathcal{E}[\overline{A}]} \mathbf{a}(e)^{-1} (\theta(e) - \theta_*(e))^2 \\
&> 2 \sum_{e \in \mathcal{E}[\overline{A}]} (\theta(e) - \theta_*(e)) \nabla^{\mathcal{G}} f(e) \qquad \text{(since } \theta_* \neq \theta\text{)} \\
&= -4 \sum_{v \in A} (\mathbf{div}(\theta - \theta_*))(v) f(v) \qquad \text{(Lemma 2.1)} \\
&= 0 \,,
\end{aligned}$$

where in the last step we used the assumption $\mathbf{div}(\theta) = \mathbf{div}(\theta_*)$ on A. □

2.2. **Geometry.** We consider the *dual polytope* associated with the edge $e = (w, v) \in \mathcal{E}$,

$$Q_e := \mathbf{conv}(w, P_w \cap P_v) \cup \mathbf{conv}(v, P_w \cap P_v) \,. \tag{2.5}$$

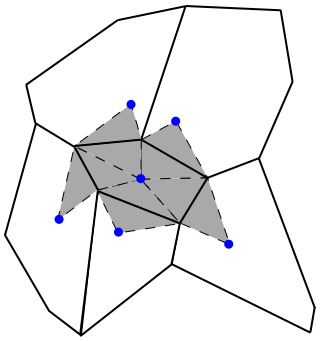


FIGURE 6. A polytope and its neighbors are outlined in thick black and the vertices are blue dots. For each neighbor, the corresponding Q_e sets are outlined in dashed lines and filled in dark gray.

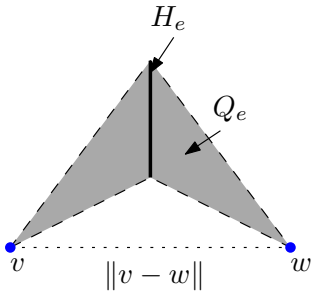


FIGURE 7. Visual aid to computing the volume of Q_e in the proof of Lemma 2.4. The points v, w are in blue, the set Q_e is in light gray outlined by dashed lines, the facet H_e is a thick black line, and the line between v and w is dotted black.

See Figures 5 and 6. This set appears in our arguments below and we will need to compute its volume. The following elementary identity may be deduced from [Droniou et al., 2018, Lemma B.2] but we include a proof for convenience.

Lemma 2.4. *For each edge $e = (w, v) \in \mathcal{E}$ we have that*

$$\|w - v\| \, \mathbf{vol}_{d-1}(H_e) = d \, \mathbf{vol}_d(Q_e) \, .$$

Proof. By an orthogonal transformation, we may assume that the facet H_e lies on the hyperplane $x_1 = 0$ and that $w = (w_1, 0, \ldots, 0)$ and $v = (-v_1, 0, \ldots, 0)$ for $v_1, w_1 > 0$. In particular, we have

$$\mathbf{conv}(w, H_e) = \{\theta x + (1-\theta)w \quad \text{such that } x \in H_e \text{ and } \theta \in [0,1]\} \, .$$

See Figure 7.

Consequently, the volume may be computed via a change of variables $\phi : \mathbb{R}^d \to \mathbb{R}^d$ where $\phi(\theta, x) = (1-\theta)x + \theta w_1$ which has Jacobian determinant $\theta^{d-1} w_1$ as follows:

$$\int_{\mathbf{conv}(w,H_e)} \mathbf{dvol}_d = \int_{\phi([0,1]\times H_e)} \mathbf{dvol}_d = \int_0^1 \int_{H_e} \theta^{d-1} w_1 \, \mathbf{dvol}_{d-1} \, d\theta = \frac{1}{d} w_1 \, \mathbf{vol}_{d-1}(H_e) \, .$$

Similarly, we have that

$$\mathbf{vol}_d(\mathbf{conv}(v, H_e)) = \frac{1}{d} v_1 \, \mathbf{vol}_{d-1}(H_e) \, .$$

The previous two displays yield the claim. □

We will also need to compare the d-dimensional volumes of sets of polytopes which intersect U.

Lemma 2.5. *Recall the definition of $\widehat{U}$ for $U \subset \mathcal{D}$ from (1.10). For every bounded set $U \subset \mathcal{D}$*

$$U \subset \bigcup_{e \in \mathcal{E}[\overline{U}]} Q_e \subset \widehat{U} \,. \tag{2.6}$$

Proof. The bound (2.6) follows immediately from the fact, by convexity, that dual polytopes cover polytopes, $P_v \subset \bigcup_{e=(w,v)\in\mathcal{E}} Q_e$, see Figure 6. □

2.3. **Convergence of the energy.** Given a subset of vertices A with $\overline{A} \subset \mathcal{V}$ and a function $f : \overline{A} \to \mathbb{R}$, the *Dirichlet energy* of f is defined as

$$\mathcal{D}(f, A) := \sum_{e \in \mathcal{E}[\overline{A}]} \mathbf{a}(e)^{-1} |\nabla^{\mathcal{G}}_{\mathbf{a}} f(e)|^2 \,. \tag{2.7}$$

When $U \subset \mathbb{R}^d$, we write

$$\mathcal{D}(f, U) := \mathcal{D}(f, \mathcal{V}[U]) \,.$$

This definition of energy may also be thought of as a discrete Sobolev norm adapted to the geometry of the orthogonal tiling graph, see, *e.g.*, [Eymard et al., 2000, Definition 9.3].

We show the energy of the difference of a continuum harmonic function and its discrete harmonic extension is small. This is well known in the finite volume literature, see for example, [Eymard et al., 2000, Theorem 9.3].

Proposition 2.6. *Let U be a bounded domain with $\overline{U} \subset \mathcal{D}$ and suppose that $h_C \in C^2(\widehat{U})$ satisfies $\mathbf{\Delta} h_C = 0$ on $\widehat{U}$ and let $h_D : \overline{\mathcal{V}[U]} \to \mathbb{R}$ solve*

$$\begin{cases} \Delta^{\mathcal{G}}_{\mathbf{a}} h_D = 0 & \text{in } \mathcal{V}[U] \,, \\ h_D = h_C & \text{on } \partial\mathcal{V}[U] \,. \end{cases}$$

Then, with

$$\varepsilon := \sup_{v \in \overline{\mathcal{V}[U]}} \operatorname{diam}(P_v) \quad \text{and} \quad M := \sup_{\widehat{U}} |\mathbf{\nabla}^2 h_C|$$

we have the bound

$$\mathcal{D}(h_D - h_C, U) \le 9 M^2 \varepsilon^2 \sum_{e \in \mathcal{E}[\overline{U}]} d\, \mathbf{vol}_d(Q_e) \,. \tag{2.8}$$

Proof. Consider the vector field $\tilde{\theta} : \mathcal{E}[\overline{U}] \to \mathbb{R}$ defined by

$$\tilde{\theta}(w, v) = \int_{P_w \cap P_v} \frac{\mathbf{\nabla} h_C(y) \cdot (w - v)}{\|w - v\|} \, \mathbf{dvol}_{d-1}(y) \qquad \forall (w, v) \in \mathcal{E}[\overline{U}] \,,$$

and observe that, by the divergence theorem, for every $v \in \mathcal{V}[U]$

$$\begin{aligned} (\mathbf{div}\, \tilde{\theta})(v) &= \sum_{(w,v)\in\mathcal{E}} \tilde{\theta}(w, v) \\ &= \sum_{(w,v)\in\mathcal{E}} \int_{P_w \cap P_v} \frac{\mathbf{\nabla} h_C(y) \cdot (w - v)}{\|w - v\|} \, \mathbf{dvol}_{d-1}(y) = \int_{P_v} \mathbf{\Delta} h_C(x) \, \mathbf{dvol}_d(x) = 0 \,. \end{aligned}$$

Define the vector field $\theta : \mathcal{E}[\overline{U}] \to \mathbb{R}$ by

$$\theta(w,v) := \tilde{\theta}(w,v) - \nabla^{\mathcal{G}}_{\mathbf{a}} h_C(w,v) \quad \forall (w,v) \in \mathcal{E}[\overline{U}]\,.$$

Let $f := h_D - h_C$ and observe that by the above two displays and the fact that h_D is discrete harmonic, we have that on $\mathcal{V}[U]$,

$$\mathbf{div}(\theta) = \mathbf{div}(-\nabla^{\mathcal{G}}_{\mathbf{a}} h_C) = \mathbf{div}(\nabla^{\mathcal{G}}_{\mathbf{a}} h_D - \nabla^{\mathcal{G}}_{\mathbf{a}} h_C) = \mathbf{div}(\nabla^{\mathcal{G}}_{\mathbf{a}} f)\,.$$

Consequently, since f is zero on $\partial\mathcal{V}[U]$, we have by Lemma 2.3 that

$$\mathcal{D}(f,U) \le \sum_{e \in \mathcal{E}[\overline{U}]} \mathbf{a}(e)^{-1} \theta(e)^2\,, \tag{2.9}$$

and so it suffices to bound the sum on the right. By definition of $\nabla^{\mathcal{G}}_{\mathbf{a}} f$ we have

$$\theta(w,v) = \int_{P_w \cap P_v} \frac{\boldsymbol{\nabla} h_C(y) \cdot (w-v) - (h_C(w) - h_C(v))}{\|w-v\|} \, \mathbf{dvol}_{d-1}(y)\,,$$

and so

$$\begin{aligned} &\mathbf{a}(w,v)^{-1}\theta(w,v)^2 \\ &= \|w-v\| \, \mathbf{vol}_{d-1}(P_w \cap P_v) \times \\ &\left(\frac{1}{\mathbf{vol}_{d-1}(P_w \cap P_v)} \int_{P_w \cap P_v} \frac{\boldsymbol{\nabla} h_C(y) \cdot (w-v) - (h_C(w) - h_C(v))}{\|w-v\|} \, \mathbf{dvol}_{d-1}(y) \right)^2 . \end{aligned} \tag{2.10}$$

By Taylor's theorem

$$|h_C(w) - (h_C(v) + \boldsymbol{\nabla} h_C(w) \cdot (w-v))| \le \frac{M}{2} \|w-v\|^2$$

and the mean value inequality,

$$|(\boldsymbol{\nabla} h_C(y) - \boldsymbol{\nabla} h_C(y')) \cdot (w-v)| \le M \, \|y-y'\| \, \|w-v\| \qquad \forall y, y' \in P_w \cup P_v\,.$$

The previous two displays and the triangle inequality yield

$$\sup_{y \in P_w \cap P_v} \frac{|\boldsymbol{\nabla} h_C(y) \cdot (w-v) - (h_C(w) - h_C(v))|}{\|w-v\|} \le 3M\varepsilon\,. \tag{2.11}$$

Combining (2.9), (2.10), and (2.11) with Lemma 2.4 completes the proof. □

3. Iterating to get a pointwise bound

In this section we prove Theorem B under Hypothesis (a). We start by converting the energy estimate (Proposition 2.6) into a pointwise bound on columns (Proposition 3.2) using a Poincaré-type bound (Lemma 3.1).

We then observe that this leads, by planarity, to a short proof when $d = 2$ (Theorem 3.3). In higher dimensions, we use Hypotheses (II) or (III) together with an iteration to prove the claims in Theorems 3.6 or 3.7 respectively.

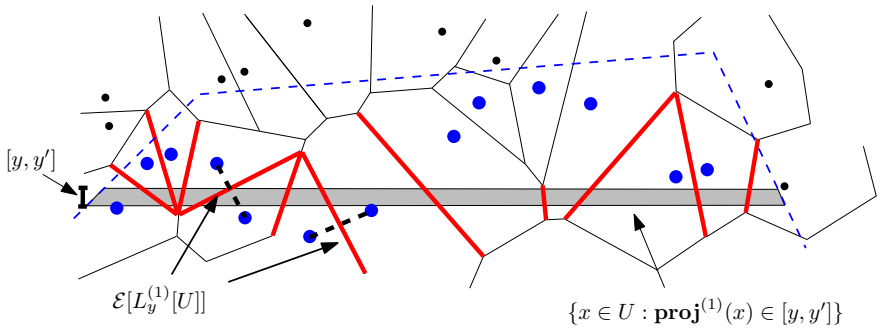


FIGURE 8. Part of Figure 4 is shown along with (red) facets which have projections intersecting a small (gray) interval $[y, y']$. Two edges in $\mathcal{E}[L_y^{(1)}[U]]$ are displayed.

3.1. **Pointwise bound on columns.** We record a bound obtained from the proof of the Poincaré inequality. The discrete Poincaré inequality which this leads to may be found in [Eymard et al., 2000, Lemma 9.1].

For $x \in \mathbb{R}^d$ and $j \in \{1, \dots, d\}$ we write $\mathbf{proj}^{(j)}(x) = (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_d)$ for the orthogonal projection of x onto $\mathbb{R}^{d-1}$ obtained by removing the jth coordinate. For a domain $U \subset \mathbb{R}^d$, $j \in \{1, \dots, d\}$, and $y \in \mathbb{R}^{d-1}$, we write

$$\begin{aligned} \mathcal{E}[L_y^{(j)}[U]] &= \{e \in \mathcal{E}[\overline{U}] : y \in \mathbf{proj}^{(j)}(H_e)\} \\ L_y^{(j)}[U] &= \{v \in \mathcal{V}[U] : v \text{ is an endpoint of } e \text{ for some } e \in \mathcal{E}[L_y^{(j)}[U]]\}\,, \end{aligned} \tag{3.1}$$

for the set of edges (resp. endpoints of edges) adjacent to U with projected facets which intersect y — see Figure 8. Note that there can be edges in $\mathcal{E}[L_y^{(j)}[U]]$ whose projections do not contain y.

Lemma 3.1 (Poincaré-type inequality)**.** *For every bounded domain U with $\overline{U} \subset \mathcal{D}$, coordinate $j \in \{1, \dots, d\}$, and function $f : \mathcal{V}[U] \to \mathbb{R}$ which is zero on $\partial\mathcal{V}[U]$ we have for every $k > 0$*

$$\mathbf{vol}_{d-1}\left(y \in \mathbf{proj}^{(j)}(U) : \max_{v \in L_y^{(j)}[U]} |f(v)| > k\left(\mathcal{D}(f, U)\sum_{e \in \mathcal{E}[\overline{U}]} d\,\mathbf{vol}_d(Q_e)\right)^{1/2}\right) \le k^{-1}\,.$$

Proof. By interchanging summation and integration,

$$\int_{\mathbf{proj}^{(j)}(U)} \sum_{e \in \mathcal{E}[L_y^{(j)}[U]]} |\nabla^{\mathcal{G}} f(e)|\,\mathbf{dvol}_{d-1}(y) \le \sum_{e \in \mathcal{E}[\overline{U}]} |\nabla^{\mathcal{G}} f(e)|\,\mathbf{vol}_{d-1}(H_e)\,.$$

Consequently, by the Cauchy–Schwarz inequality,

$$\begin{aligned}
&\int_{\mathbf{proj}^{(j)}(U)} \sum_{e\in\mathcal{E}[L_y^{(j)}[U]]} |\nabla^{\mathcal{G}} f(e)|\,\mathbf{dvol}_{d-1}(y)\\
&\le \sqrt{\sum_{e\in\mathcal{E}[\overline{U}]} |\nabla^{\mathcal{G}} f(e)|^2 \mathbf{a}(e)}\sqrt{\sum_{e\in\mathcal{E}[\overline{U}]} \mathbf{a}^{-1}(e)\,\mathbf{vol}_{d-1}(H_e)^2}\\
&= \sqrt{\sum_{e\in\mathcal{E}[\overline{U}]} |\nabla^{\mathcal{G}} f(e)|^2 \mathbf{a}(e)}\sqrt{\sum_{e\in\mathcal{E}[\overline{U}]} d\,\mathbf{vol}_d(Q_e)}\,,
\end{aligned}$$

where in the last equality we used Lemma 2.4. This implies

$$\int_{\mathbf{proj}^{(j)}(U)} \sum_{e\in\mathcal{E}[L_y^{(j)}[U]]} |\nabla^{\mathcal{G}} f(e)|\,\mathbf{dvol}_{d-1}(y) \le \mathcal{D}(f,U)^{1/2}\left(\sum_{e\in\mathcal{E}[\overline{U}]} d\,\mathbf{vol}_d(Q_e)\right)^{1/2}. \tag{3.2}$$

Observe that for $\mathbf{vol}_{d-1}$-almost every $y\in\mathbf{proj}^{(j)}(U)$, since f is zero on the boundary, by summing the gradient,

$$\sum_{e\in\mathcal{E}[L_y^{(j)}[U]]} |\nabla^{\mathcal{G}} f(e)| < \delta \implies \max_{v\in L_y^{(j)}[U]} |f(v)| < \delta \qquad \forall \delta>0\,.$$

Here we are using the fact that for $\mathbf{vol}_{d-1}$-almost every $y\in\mathbf{proj}^{(j)}(U)$ the vertices in $L_y^{(j)}[U]$ are connected via the edges in $\mathcal{E}[L_y^{(j)}[U]]$.

By the above two displays and Chebyshev's inequality, we have that, for all $\delta>0$

$$\mathbf{vol}_{d-1}\left(y\in\mathbf{proj}^{(j)}(U) : \max_{v\in L_y^{(j)}[U]} |f(v)| > \delta\right) \le \delta^{-1}\mathcal{D}(f,U)^{1/2}\left(\sum_{e\in\mathcal{E}[\overline{U}]} d\,\mathbf{vol}_d(Q_e)\right)^{1/2}.$$

This completes the proof. □

By combining the above inequality together with Proposition 2.6 we obtain a pointwise bound on the difference between the discrete and continuous harmonic functions on an arbitrarily dense set of columns.

Proposition 3.2 (Closeness on columns)**.** *Let U be a bounded domain with $\overline{U}\subset\mathcal{D}$ and suppose that $h_C\in C^2(\widehat{U})$ satisfies $\mathbf{\Delta} h_C = 0$ on $\widehat{U}$ and let $h_D : \overline{\mathcal{V}[U]}\to\mathbb{R}$ satisfy*

$$\begin{cases} \Delta_{\mathbf{a}}^{\mathcal{G}} h_D = 0 & \text{in } \mathcal{V}[U]\,,\\ h_D = h_C & \text{on } \partial\mathcal{V}[U]\,. \end{cases}$$

Then, with

$$\varepsilon := \sup_{v\in\overline{\mathcal{V}[U]}} \operatorname{diam}(P_v) \quad \text{and} \quad M := \sup_{\widehat{U}} |\mathbf{\nabla}^2 h_C|$$

we have for every $k>0$ and $j\in\{1,\dots,d\}$,

$$\mathbf{vol}_{d-1}\left(y\in\mathbf{proj}^{(j)}(U) : \max_{v\in L_y^{(j)}[U]} |h_D(v)-h_C(v)| > 3kM\varepsilon \sum_{e\in\mathcal{E}[\overline{U}]} d\,\mathbf{vol}_d(Q_e)\right) \le k^{-1}.$$

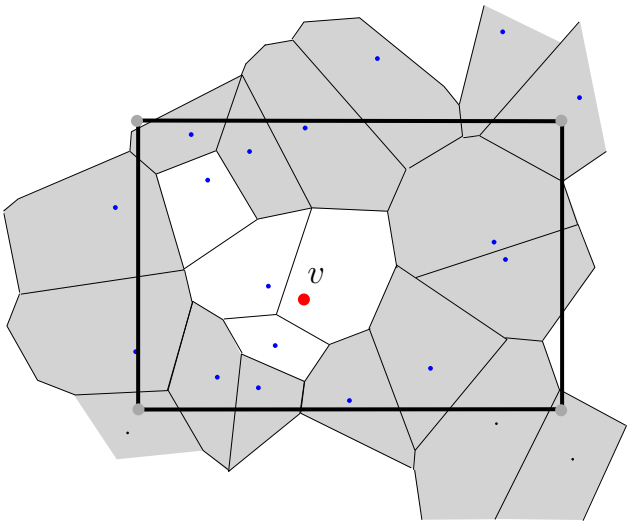


FIGURE 9. Illustration of the proof of Theorem 3.3. The square S is outlined in black, the cells for which the discrete harmonic function is close to the continuum one are in light gray.

Proof. This is immediate from Lemma 3.1 and Proposition 2.6. □

3.2. **Two dimensions.** We give a short proof of a strengthened version of the main result in Gurel-Gurevich et al. [2020]. Roughly, when $d = 2$, by taking k large enough, Proposition 3.2 guarantees that there is a fine grid of "good lines" upon which the discrete and continuous harmonic functions h_D and h_C are close. On every rectangle formed by the grid, by smoothness, h_C is essentially constant. Thus, by the maximum principle, h_D is close to h_C everywhere.

Theorem 3.3. *Let U be a bounded domain with $\overline{U} \subset \mathcal{D} \subset \mathbb{R}^2$ and suppose $h_C \in C^2(\widehat{U})$ satisfies $\mathbf{\Delta} h_C = 0$ on $\widehat{U}$ and let $h_D : \overline{\mathcal{V}[U]} \to \mathbb{R}$ satisfy*

$$\begin{cases} \Delta_{\mathbf{a}}^{\mathcal{G}} h_D = 0 & \text{in } \mathcal{V}[U]\,, \\ h_D = h_C & \text{on } \partial\mathcal{V}[U]\,. \end{cases}$$

Then, with

$$\varepsilon := \sup_{v \in \overline{\mathcal{V}[U]}} \operatorname{diam}(P_v) \quad L := \sup_{\widehat{U}} |\boldsymbol{\nabla} h_C| \quad \text{and} \quad M := \sup_{\widehat{U}} |\boldsymbol{\nabla}^2 h_C|\,,$$

we have the pointwise bound

$$\sup_{v \in \mathcal{V}[U]} |h_D(v) - h_C(v)| \le \inf_{k>0} \left(6kM\varepsilon\, \mathbf{vol}_d(\widehat{U}) + 2(\sqrt{2}k^{-1} + \varepsilon)L \right)\,.$$

In particular, there exists $C = C(U, M, L) < \infty$ so that

$$\sup_{v \in \mathcal{V}[U]} |h_D(v) - h_C(v)| \le C\varepsilon^{1/2}\,.$$

Proof. By Proposition 3.2 we have, for each $j \in \{1,2\}$ and every $k > 0$,

$$\left| \underbrace{\left\{ y \in \mathbf{proj}^{(j)}(U) : \max_{v \in L_y^{(j)}[U]} (h_D(v) - h_C(v)) > 6kM\varepsilon \, \mathbf{vol}_d(\widehat{U}) \right\}}_{=:Y^{(j)}} \right| \leq k^{-1}\,, \tag{3.3}$$

where we also used (2.6). Let $v \in \mathcal{V}(U)$ be given and suppose $h_D(v) - h_C(v) > 6kM\varepsilon \, \mathbf{vol}_d(\widehat{U})$.

We claim that the above display implies that there exists a rectangle S, with $v \in S$, such that

$$\operatorname{diam}(S) \leq 2\sqrt{2}k^{-1} \quad \text{and} \quad \max_{w \in \partial\mathcal{V}[S \cap U]} (h_D(w) - h_C(w)) < 6kM\varepsilon \, \mathbf{vol}_d(\widehat{U}) \quad \forall k > 0\,, \tag{3.4}$$

see Figure 9. Indeed, let $S = S^{(2)} \times S^{(1)}$ where $S^{(j)}$ is the largest interval in $Y^{(j)}$ containing v_i. By (3.3), $\operatorname{diam}(S) \leq 2\sqrt{2}k^{-1}$. Also, by maximality of each interval, if $w \in \partial\mathcal{V}[S \cap U] \setminus \partial\mathcal{V}[U]$, then $(h_D(w) - h_C(w)) < 6kM\varepsilon \, \mathbf{vol}_d(\widehat{U})$. If $w \in \partial\mathcal{V}[U]$, then $h_D(w) - h_C(w) = 0$. This establishes (3.4).

We also have by the mean value theorem that

$$\sup_{x,y \in \overline{\mathcal{V}[S \cap U]}} |h_C(x) - h_C(y)| \leq (\operatorname{diam}(S) + 2\varepsilon)L\,.$$

Since $v \in S$, by the maximum principle $h_D(v) \leq h_D(w)$ for some vertex $w \in \partial\mathcal{V}[S \cap U]$. Consequently, by the above display, (3.4) and the triangle inequality we have, for all $k > 0$,

$$\begin{aligned} h_D(v) - h_C(v) &\leq h_D(w) - h_C(v) \\ &\leq h_D(w) - h_C(w) + 2(\sqrt{2}k^{-1} + \varepsilon)L \\ &\leq 6kM\varepsilon \, \mathbf{vol}_d(\widehat{U}) + 2(\sqrt{2}k^{-1} + \varepsilon)L\,. \end{aligned}$$

A symmetric argument establishes the bound in the other direction. The "in particular" clause follows by choosing $k = \varepsilon^{-1/2}$. □

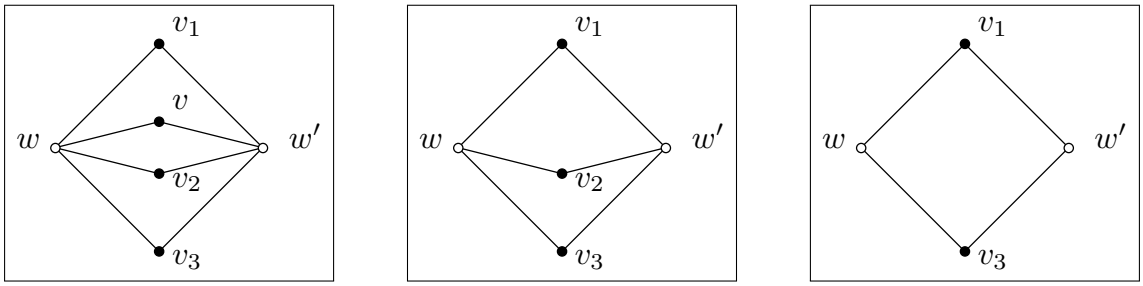


FIGURE 10. Series reduction of an orthodiagonal map; from left to right the vertices v and v_2 in $\mathcal{V}^\bullet$ of degree two are removed. The same labeling scheme as Figure 3 is used.

Remark 3.4. *Recall the definition of orthodiagonal map from Section 1.2.3. In the case some quadrilaterals in $\mathcal{G}$ are nonconvex, the construction there may fail to produce an orthogonal tiling. In this remark, we explain how our proof still applies to this case.*

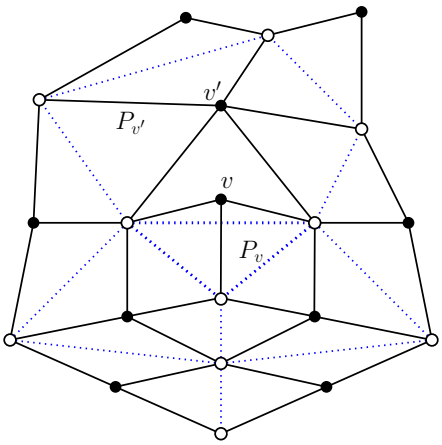


FIGURE 11. A vertex v in an orthodiagonal map which is not contained in its corresponding polytope P_v but contained in a polytope of a neighbor $P_{v'}$. The same labeling scheme as Figure 3 is used.

We first observe that we can use electric network laws to reduce to the case where all interior vertices in $\mathcal{G}^\bullet$ and $\mathcal{G}^\circ$ have degree at least three. An interior vertex in $\mathcal{G}$ must be incident to at least two inner faces, so every interior vertex in $\mathcal{G}^\bullet$ has degree at least two. An interior vertex v in $\mathcal{G}^\bullet$ with exactly two neighbors $v_1, v_2 \in \mathcal{G}^\bullet$ must share an edge in $\mathcal{E}$ with exactly two vertices $w, w' \in \mathcal{V}^\circ$. Thus, by the series law, any function $f : \mathcal{V}^\bullet \to \mathbb{R}$ which is $\mathbf{a}$-discrete harmonic must also be $\mathbf{a}$-discrete harmonic on the graph where the vertex v and its two incident edges in $\mathcal{G}$ is removed — see Figure 10. Indeed, since v_1 and v_2 are neighbors of v and $w - w'$ is orthogonal to $v - v_1$ and $v - v_2$, the vertex v lies on the line segment between v_1 and v_2; therefore

$$\frac{\|v_1 - v_2\|}{\|w - w'\|} = \frac{\|v_1 - v\| + \|v - v_2\|}{\|w - w'\|} = \frac{1}{\mathbf{a}(v_1, v)} + \frac{1}{\mathbf{a}(v_2, v)}\,.$$

Similarly, by the parallel law, we may reduce to the case there are no internal vertices in V° of degree less than three. The result of Theorem 3.3 on the reduced graph implies the result on the non-reduced graph. Indeed, arguing with the maximum principle for discrete harmonic functions and the mean-value property as in the proof of Theorem 3.3, if a discrete harmonic function is close to a continuum harmonic function on the reduced graph, then it must be close to the same continuum harmonic function on the deleted vertices.

Every vertex $v \in \mathcal{V}^\bullet$ is now associated to a well-defined strictly convex polygon P_v with edges given by the opposite diagonals of the quadrilaterals containing v. The polygons $\{P_v\}_{v \in \mathcal{V}^\bullet}$ have pairwise disjoint interiors. Every vertex $v \in \mathcal{V}^\bullet$ is either contained in the interior of P_v or in the interior of $P_{v'}$ where $v' \in \mathcal{V}^\bullet$ is a neighbor of v — see Figure 11.

We also define, for each edge $e = (v, v') \in \mathcal{E}^\bullet$, the set Q_e to be the quadrilateral in $\mathcal{G}$ with extremal vertices v and v'. Note for convex quadrilaterals, this coincides with Q_e defined in (2.5).

With these two modified definitions, the proof of Theorem 3.3 proceeds in the exact same way.

3.3. **All dimensions with smallness hypotheses.** We now iterate Proposition 3.2 together with an assumption on tiny polytopes to go from smallness on columns to a pointwise

bound. The following bound on the measure of dual polytopes which correspond to an edge with large gradient will be used.

Lemma 3.5 (Weighted gradient-energy estimate)**.** *For every bounded set of vertices A with $\overline{A} \subset \mathcal{V}$ and function $f : \overline{A} \to \mathbb{R}$ we have the bound*

$$\sum_{e \in \mathcal{E}[\overline{A}]} d\,\mathbf{vol}_d(Q_e) 1\{|\nabla^{\mathcal{G}} f(e)| > \delta\} \le \mathcal{D}(f, A) \qquad \forall \delta \ge \sup_{e \in \mathcal{E}[\overline{A}]} \|e\| .$$

Proof. We compute, using the assumption $\delta \ge \sup_{e \in \mathcal{E}[\overline{A}]} \|e\|$,

$$\begin{aligned} \mathcal{D}(f, A) = \sum_{e \in \mathcal{E}[\overline{A}]} \frac{\mathbf{vol}_{d-1}(H_e)}{\|e\|} |\nabla^{\mathcal{G}} f(e)|^2 &\ge \sum_{e \in \mathcal{E}[\overline{A}]} \frac{\mathbf{vol}_{d-1}(H_e)|}{\|e\|} \delta^2 1\{|\nabla^{\mathcal{G}} f(e)| > \delta\} \\ &\ge \sum_{e \in \mathcal{E}[\overline{A}]} \mathbf{vol}_{d-1}(H_e)\|e\| 1\{|\nabla^{\mathcal{G}} f(e)| > \delta\} \\ &= \sum_{e \in \mathcal{E}[\overline{A}]} d\,\mathbf{vol}_d(Q_e) 1\{|\nabla^{\mathcal{G}} f(e)| > \delta\} , \end{aligned}$$

where in the last step we used Lemma 2.4. □

Theorem 3.6. *Let U be a bounded domain with $\overline{U} \subset \mathcal{D} \subset \mathbb{R}^d$ and suppose $h_C \in C^2(\widehat{U})$ satisfies $\mathbf{\Delta} h_C = 0$ on $\widehat{U}$ and let $h_D : \overline{\mathcal{V}[U]} \to \mathbb{R}$ satisfy*

$$\begin{cases} \Delta^{\mathcal{G}}_{\mathbf{a}} h_D = 0 & \text{in } \mathcal{V}[U] , \\ h_D = h_C & \text{on } \partial\mathcal{V}[U] . \end{cases}$$

Then, there exists $C = C(U) < \infty$ such that, with

$$\varepsilon := \sup_{v \in \overline{\mathcal{V}[U]}} \operatorname{diam}(P_v) \quad L := \sup_{\widehat{U}} |\mathbf{\nabla} h_C| \quad \text{and} \quad M := \sup_{\widehat{U}} |\mathbf{\nabla}^2 h_C| \vee 1 ,$$

and for any fixed $K \ge 1$, which can depend on ε,

$$\mathcal{A} := \{v \in \overline{\mathcal{V}[U]} : \mathbf{vol}_d(P_v) < \varepsilon^K\} \quad \text{and} \quad D := \sup_{\text{components } Q \subset \mathcal{A}} \sum_{v \in Q} \operatorname{diam}(P_v) , \tag{3.5}$$

where the supremum is over the connected components of the subgraph induced by $\mathcal{A}$, we have, assuming $\varepsilon < (CM)^{-1}$,

$$\sup_{v \in \mathcal{V}[U]} |h_D(v) - h_C(v)| \le CMK\varepsilon(\log \varepsilon^{-1}) + DL . \tag{3.6}$$

Note that if for some large K the set $\mathcal{A}$ in (3.5) is empty, then the second error term in (3.6) is zero.

Proof of Theorem 3.6. We consider a sequence of functions $\{h_i\}_{i \ge 0}$ on $\mathcal{V}[U]$ which converges to h_C as $i \to \infty$. Each such function will be the "discrete harmonic extension" of h_C to a subset $S_i \subset \mathcal{V}[U]$. In the first step we define the sequence. In the second step we bound $|h_D - h_C|$ in terms of $|h_i - h_D|$ and the diameter of the components of S_i using the maximum principle and the continuity of h_C. In the third step, we estimate this diameter using (3.5) and Proposition 3.2.

Step 1. We define the iteration. For a set of vertices V with $V \subset \mathcal{V}[U]$ we denote the harmonic extension of h_C to V by $\mathcal{H}(V)$, *i.e.*, $\mathcal{H}(V) = g$ where g is the unique solution $g : \overline{V} \to \mathbb{R}$ of

$$\begin{cases} \Delta_{\mathbf{a}}^{\mathcal{G}} g = 0 & \text{in } V\,, \\ g = h_C & \text{on } \overline{\mathcal{V}[U]} \setminus V\,. \end{cases}$$

Fix a large constant $A > 1$, depending on U, to be determined below. Start with $h_0 := h_D$ and recursively set, for $i \geq 0$,

$$h_{i+1} = \mathcal{H}(S_i)$$

where $S_0 := \mathcal{V}[U]$ and

$$\begin{aligned} N_{i+1} &:= \{v \in \mathcal{V}[U] : |(h_i - h_C)(v)| > AM\varepsilon\}\,, \\ S_{i+1} &:= N_{i+1} \setminus \{v \in N_{i+1} : \exists w \in \partial N_{i+1} \text{ with } |\nabla^{\mathcal{G}}(h_i - h_C)(v,w)| \leq 2\varepsilon\}\,. \end{aligned} \tag{3.7}$$

That is, S_{i+1} is defined to be a slightly trimmed subset of N_{i+1}, which itself is a subset of S_i where $|h_i - h_C|$ is large. The trimming is used in (3.10) below so that Lemma 3.5 can control the volume of polytopes on the boundary of N_{i+1} which remain. Note that $h_0 = h_1 = h_D$.

Step 2. We bound the difference between h_C and h_D using the sequence defined in Step 1. By definition of S_i, we have that

$$|h_i - h_C| \leq (AM+2)\varepsilon \quad \text{on } \mathcal{V}[U] \setminus S_{i+1} \qquad \forall i \geq 0\,.$$

Also, since the sets are necessarily nested, $S_{i+1} \subset S_i$, the maximum principle yields that for every $i \geq 0$,

$$\max_{\mathcal{V}[U]} |h_{i+1} - h_i| = \max_{S_i} |h_{i+1} - h_i| = \max_{\partial S_i} |h_{i+1} - h_i| = \max_{\partial S_i} |h_C - h_i| \leq (AM+2)\varepsilon\,.$$

By the previous two displays and telescoping, we have

$$|h_D - h_C| \leq i(AM+2)\varepsilon \quad \text{on } \overline{\mathcal{V}[U]} \setminus S_i \quad \forall i \geq 0\,. \tag{3.8}$$

By the mean value theorem, for each component Q of S_i,

$$\sup_{x,y \in \overline{Q}} |h_C(x) - h_C(y)| \leq \sum_{v \in Q} \operatorname{diam}(P_v) L$$

and, for each vertex $v \in Q$, by the maximum principle $h_D(v) \leq h_D(w)$ for some vertex $w \in \partial Q$. Consequently, by the above two displays and the triangle inequality, for each such component Q and vertex $v \in Q$, there is a vertex $w \in \partial Q$ such that

$$\begin{aligned} h_D(v) - h_C(v) &\leq h_D(w) - h_C(v) \\ &\leq h_D(w) - h_C(w) + \sum_{v \in Q} \operatorname{diam}(P_v) L \\ &\leq i(AM+2)\varepsilon + \sum_{v \in Q} \operatorname{diam}(P_v) L\,. \end{aligned}$$

Repeating this for the other direction shows that

$$\sup_{v \in \mathcal{V}[U]} |h_D(v) - h_C(v)| \leq i(AM+2)\varepsilon + \left(\sup_{\text{components } Q \subset S_i} \sum_{v \in Q} \operatorname{diam}(P_v) \right) L \quad \forall i \geq 0\,. \tag{3.9}$$

Step 3. We show that the term in the parentheses on the right of (3.9) is bounded above by D for a sufficiently large choice of i. For each $i \geq 0$, let

$$U_i := \bigcup_{v \in S_i} P_v \quad \text{and} \quad Y_i := \{y \in \mathbf{proj}^{(1)}(U_i) : \max_{v \in L_y^{(1)}[U_i]} |(h_i - h_C)(v)| > AM\varepsilon\},$$

where $\mathbf{proj}^{(1)}$ and $L_y^{(1)}$ are as in (3.1). By Proposition 3.2 with parameter choice of $k = (3A^{-1}\sum_{e \in \mathcal{E}[\overline{U_i}]} d\,\mathbf{vol}_d(Q_e))^{-1}$ and U replaced by U_i we have, for every $i \geq 0$,

$$\mathbf{vol}_{d-1}(Y_i) \leq 3A^{-1} \sum_{e \in \mathcal{E}[\overline{U_i}]} d\,\mathbf{vol}_d(Q_e).$$

Using the definitions of U_i and S_i, we may bound the sum on the right as

(3.10)
$$\sum_{e \in \mathcal{E}[\overline{U}_i]} d\,\mathbf{vol}_d(Q_e) \leq \underbrace{\sum_{e \in \mathcal{E}[N_i]} d\,\mathbf{vol}_d(Q_e)}_{\leq d\,\mathbf{vol}_{d-1}(Y_{i-1})\,\mathrm{diam}(U)} + \underbrace{\sum_{e \in \partial\mathcal{E}[N_i]} d\,\mathbf{vol}_d(Q_e) 1\{|\nabla^{\mathcal{G}}(h_{i-1} - h_C)(e)| > 2\varepsilon\}}_{\leq \mathcal{D}(h_{i-1} - h_C, U_{i-1})}.$$

The bound for the first term follows since, for each edge $(w, v) = e \in \mathcal{E}[N_i]$, as $v, w \in N_i$, the projection of $P_w \cup P_v$ is contained in Y_{i-1} and hence the dual polytope Q_e is completely contained in a cylinder with base Y_{i-1} and height $\mathrm{diam}(U)$. The second term is bounded by Lemma 3.5 applied with $f := h_{i-1} - h_C$ and $A = U_{i-1}$.

Combining the above two displays with Proposition 2.6 to bound $\mathcal{D}(h_i - h_C, U_i)$ yields, for every $i \geq 0$

$$\begin{aligned}\mathbf{vol}_{d-1}(Y_{i+1}) &\leq 3A^{-1} \sum_{e \in \mathcal{E}[\overline{U_{i+1}}]} d\,\mathbf{vol}_d(Q_e) \\ &\leq 3A^{-1} d\,\mathrm{diam}(U)\,\mathbf{vol}_{d-1}(Y_i) + 27A^{-1}M^2\varepsilon^2 \sum_{e \in \mathcal{E}[\overline{U_i}]} d\,\mathbf{vol}_d(Q_e),\end{aligned}$$

and so, if we define

$$f(i) := \mathbf{vol}_{d-1}(Y_i) \vee 3A^{-1} \sum_{e \in \mathcal{E}[\overline{U_i}]} d\,\mathbf{vol}_d(Q_e)$$

we have

$$f(i+1) \leq 3A^{-1} d\,\mathrm{diam}(U) f(i) + 9M^2\varepsilon^2 f(i).$$

Assuming $\varepsilon < 2^{-10}A^{-1}M^{-1}$, the above display implies that there is a constant $C = C(U) < \infty$ such that

$$f(i+1) \leq CA^{-1} f(i) \qquad \forall i \geq 1$$

and so, using $f(0) \leq C$, and picking A sufficiently large, depending on U, so that $CA^{-1} < \frac{1}{4}$, we have

(3.11)
$$f(i+1) \leq 2^{-i} \qquad \forall i \geq 1.$$

Since, for every $v \in S_i$, the polytope $P_v \subset \bigcup_{e \in \mathcal{E}[U_i]} Q_e$, this shows that

$$\mathbf{vol}_d(P_v) \leq 2^{-i} \qquad \forall v \in S_{i+1}, \quad \forall i \geq 1.$$

Consequently, by (3.5) each component of $S_{CK \log \varepsilon^{-1}}$ must have the sum of the diameters of its corresponding polytopes bounded by D. This together with (3.9) completes the proof. □

We modify the previous proof to replace the assumption (3.5) by one where the diameter of every polytope is bounded by some (possibly tiny) power of the volume of its dual polytopes.

Theorem 3.7. *Let U be a bounded domain with $\overline{U} \subset \mathcal{D} \subset \mathbb{R}^d$ and suppose $h_C \in C^2(\widehat{U})$ satisfies $\mathbf{\Delta} h_C = 0$ on $\widehat{U}$ and let $h_D : \overline{\mathcal{V}[U]} \to \mathbb{R}$ satisfy*

$$\begin{cases} \Delta_{\mathbf{a}}^{\mathcal{G}} h_D = 0 & \text{in } \mathcal{V}[U]\,, \\ h_D = h_C & \text{on } \partial\mathcal{V}[U]\,. \end{cases}$$

Then with

$$\varepsilon := \sup_{v \in \overline{\mathcal{V}[U]}} \operatorname{diam}(P_v) \qquad M := \sup_{\widehat{U}} |\nabla^2 h_C| \vee 1\,,$$

and under the assumption that there exists some $F, \alpha > 0$ such that

$$\operatorname{diam}(P_v) \le F \sup_{e=(w,v)\in\mathcal{E}} \mathbf{vol}_d(Q_e)^\alpha \qquad \forall v \in \mathcal{V}[U]\,, \tag{3.12}$$

we have that there exists $C = C(U, F, \alpha) < \infty$ such that, assuming $\varepsilon < (CM)^{-1}$,

$$\sup_{v\in\mathcal{V}[U]} |h_D(v) - h_C(v)| \le CM\varepsilon(\log \varepsilon^{-1})\,. \tag{3.13}$$

Proof. The proof follows the same outline as the previous one but utilizes the observation that the diameter of each polytope in the "bad set" should decrease in every step. To avoid repetition, we use the same notation as the previous proof. Objects which are nearly identical to those in the previous proof are denoted with a prime, those which are new have different letters; a central role will be played by the decreasing sequence $\{\delta_i\}_{i\ge 0}$ which will describe the longest edge in the bad set. The constant C below will be allowed to depend on U, F, α and may change from line to line. Fix $A \ge 1$, which depends only on U, as in (3.7).

Start with $h'_0 := h_D$, $\delta_0 := \varepsilon$, $S'_0 := \mathcal{V}[U]$ and recursively set, for $i \ge 0$,

$$\begin{aligned} h'_{i+1} &:= \mathcal{H}(S'_i)\,, \\ N'_{i+1} &:= \{v \in \mathcal{V}[U] : |(h'_i - h_C)(v)| > AM\delta_i\}\,, \\ S'_{i+1} &:= N'_{i+1} \setminus \{v \in N'_{i+1} : \exists w \in \partial N'_{i+1} \text{ with } |\nabla^{\mathcal{G}}(h'_i - h_C)(v,w)| \le 2\delta_i\}\,, \\ \delta_{i+1} &:= \sup_{v\in S'_{i+1}} \operatorname{diam}(P_v)\,. \end{aligned} \tag{3.14}$$

Following the argument leading to (3.8), we have

$$|h_D - h_C| \le CAM \sum_{j=1}^{i} \delta_j \quad \text{on } \overline{\mathcal{V}[U]} \setminus S'_i \quad \forall i \ge 0\,. \tag{3.15}$$

Now, instead of using (3.9) and arguing that the components of S'_i have a small diameter for i large, we will allow i to go to infinity in the above display and control how quickly δ_j goes to zero.

Since δ_j is defined as the smallest diameter of a polytope in S'_j, we may replace every instance of ε in Step 3 of the previous proof by δ_j. Denote by $U'_j := \bigcup_{v\in S'_j} P_v$ so that if we define, in place of $f(i)$,

$$f'(i) := CA^{-1} \sum_{e\in\mathcal{E}[\overline{U'_i}]} d\,\mathbf{vol}_d(Q_e)\,,$$

we have, as in (3.11),

$$f'(i+1) \le 2^{-i}\,.$$

We rewrite the bound on the sum as a bound on the maximum,

$$(CAf'(i))^\alpha \ge \left(\sum_{e\in\mathcal{E}[\overline{U_i'}]} \mathbf{vol}_d(Q_e)\right)^\alpha \ge \sup_{e\in\mathcal{E}[\overline{U_i'}]} \mathbf{vol}_d(Q_e)^\alpha\,.$$

The above two displays together with (3.12) then imply a bound on δ_i,

$$\sup_{v\in S_i'} \operatorname{diam}(P_v) \le CA^\alpha f'(i)^\alpha \le C2^{-\alpha i}\,,$$

where in the last step we absorbed A into C.

Since $\delta_i \le \varepsilon$ for all i, this shows that

$$\delta_i \le \varepsilon \wedge (C2^{-\alpha i}) \quad \forall i \ge 0\,.$$

Since the right side goes to zero as $i\to\infty$, and there are only finitely many polytopes in U, the set S_i' must empty for large enough i. Thus, taking $K\in\mathbb{N}$ and plugging the above display into (3.15) and sending $i\to\infty$ yields

$$\begin{aligned}
|h_D - h_C| &\le \inf_{K>0}\left(CMK\varepsilon + CM\sum_{j=K}^{\infty} 2^{-\alpha j}\right)\\
&\le \inf_{K>0}\left(CMK\varepsilon + CM2^{-\alpha K}\right)\\
&\le CM\varepsilon(\log\varepsilon^{-1}) \quad \text{(after choosing } K = C(\alpha^{-1}+1)\log(\varepsilon^{-1}))\,.
\end{aligned}$$

This completes the proof. □

Proof of Theorem B under Hypothesis (a). The assertions corresponding to the hypotheses (I) and (III) follow from Theorems 3.3 and 3.7 respectively. The assertion corresponding to Hypothesis (II) follows by applying Theorem 3.6 for each n with $K := 2\lfloor \frac{\log(1/\min_{v\in\mathcal{V}_n}\mathbf{vol}_d(P_v))}{\log(1/\varepsilon_n)}\rfloor$. Indeed, with this choice, for large enough n the set $\mathcal{A}$ defined in (3.5) is empty but the first term on the right in (3.6) still converges to zero. □

4. Convergence of random walk modulo time paramaterization

In this section we prove a "black box theorem" which roughly states that if you have convergence of the Dirichlet problem, then the trace of random walk converges, under rescaling, to the trace of Brownian motion. Slightly more precisely, we show that under this condition, the law of the rescaled random walk converges to the law of Brownian motion with respect to the local topology on curves viewed modulo time parameterization.

Specifically, we fix a sequence of undirected graphs $\{\mathcal{G}_n = (\mathcal{V}_n, \mathcal{E}_n)\}_{n\ge 0}$ with each $\mathcal{V}_n \subset \mathbb{R}^d$. For a point $z\in\mathbb{R}^d$, let z^n be the nearest vertex in $\mathcal{G}_n$, with ties broken in lexicographical ordering. We also fix a nearest neighbor discrete-time random walk $\{X_t^n\}_{t\ge 0}$ on $\mathcal{V}_n$; that is, $\{X_t^n\}$ is a time homogeneous Markov chain and $(X_t^n, X_{t+1}^n)\in\mathcal{E}_n$ for all $t\ge 0$. We let $\Delta^{\mathcal{G}_n}$ denote the generator of X_t^n, that is, for every vertex $a\in\mathcal{V}_n$ and function $f:\mathcal{V}_n\to\mathbb{R}$,

$$\Delta^{\mathcal{G}_n} f(a) = \mathbb{E}[f(X_1^{a,n}) - f(a)]\,,$$

where the expected value is taken over one step of the random walk with initial point $a\in\mathcal{V}_n$.

We define, for a set $U \subset \mathbb{R}^d$ and $r > 0$,

$$B_r(U) := \{x \in \mathbb{R}^d : \inf_{y \in U} \|x - y\| \leq r\}.$$

The following assumption states that the graph approximates some (possibly infinite) domain $\mathcal{D}$ and the Dirichlet problem for every ball $B_r(x)$ intersecting a neighborhood of $\mathcal{D}$ converges.

Assumption D. *There exists a domain $\mathcal{D}$ and some $\delta > 0$ such that for every $r > 0$ and $x \in B_\delta(\mathcal{D})$ the following holds for $B := B_r(x) \cap B_\delta(\mathcal{D})$. The graph approximates Euclidean space in the ball,*

$$\lim_{n \to \infty} \sup_{z \in B} \|z^n - z\| = 0. \tag{4.1}$$

Moreover, for every function h_C which is continuum harmonic in a neighborhood of B, the discrete harmonic extension h_D^n of h_C to $\mathcal{V}_n[B]$, defined by,

$$\begin{cases} \Delta^{\mathcal{G}_n} h_D^n = 0 & \text{in } \mathcal{V}_n[B], \\ h_D^n = h_C & \text{on } \partial \mathcal{V}_n[B], \end{cases}$$

converges uniformly to h_C,

$$\lim_{n \to \infty} \sup_{z \in \mathcal{V}_n[B]} |h_D^n(z) - h_C(z)| = 0. \tag{4.2}$$

Under this assumption, we prove a central limit theorem.

Theorem 4.1. *Suppose $\mathcal{D}$ is such that Assumption D holds and let $U \subset \mathcal{D}$ be a Lipschitz, bounded domain. Let $z \in U$, and let $X^{z,n}$ be random walk on $\mathcal{G}_n$ started at z^n, let $J_U^{z,n}$ denote the first exit from $\mathcal{V}_n[U]$, and let $Y^{z,n}$ denote the piecewise linear interpolation of $X^{z,n}$. Let $\mathcal{B}^z$ be a standard Brownian motion started at $z \in U$ with first exit τ_U^z from U. The supremum over all $z \in U$ of the Prokhorov distance between the laws of $Y^{z,n}|_{[0, J_U^{z,n}]}$ and $\mathcal{B}^z|_{[0, \tau_U^z]}$ with respect to the metric (1.2) converges to 0 as $n \to \infty$.*

Proof of Theorem A assuming Theorem 4.1. It follows from Theorem B that Assumption D is satisfied. This immediately implies Theorem A. □

Proof of Theorem B assuming Hypothesis (b). This is immediate from Theorem 4.1 and the fact that the solution of the Dirichlet problem can be expressed in terms of the random walk. □

It remains to prove Theorem 4.1. We first show that Assumption D implies convergence of harmonic measure. We then use the strong Markov property of random walk together with an approximation argument to deduce Theorem 4.1.

4.1. **Convergence of harmonic measure.** Let $\mathbb{B}$ denote the unit ball in $\mathbb{R}^d$. Let $\mathrm{hm}(x)$ be the harmonic measure on $\partial\mathbb{B}$, as seen from $x \in \mathbb{B}$. For $x \in \mathbb{B}$ and $z \in \partial \mathcal{V}_n[\mathbb{B}]$, we define $\mathrm{hm}^{\mathrm{n}}(x)[z]$ to be the probability measure which assigns mass to each vertex of $\partial \mathcal{V}_n[\mathbb{B}]$ equal to its discrete harmonic measure from x^n. We also write d^{Pr} for the Prokohorov distance between two probability measures.

Proposition 4.2. *Under Assumption D we have that*

$$\lim_{n \to \infty} \sup_{x \in \mathbb{B}} \mathrm{d}^{\mathrm{Pr}}(\mathrm{hm}^{\mathrm{n}}(x), \mathrm{hm}(x)) = 0.$$

Proof. Let f be a smooth, bounded function on $\partial\mathbb{B}$. By Whitney extension, we may assume that f is smooth on $\mathbb{R}^d$. It suffices to show that

$$\lim_{n\to\infty} \sup_{x\in\mathbb{B}} \left| \int_{\mathbb{R}^d} f(z) d\,\mathrm{hm}^{\mathrm{n}}(x)[z] - \int_{\partial\mathbb{B}} f(z) d\,\mathrm{hm}(x)[z] \right| = 0\,. \tag{4.3}$$

Fix $\delta > 0$ and let h_C^δ denote the unique function which is harmonic on $B_{1+\delta}$ with $h_C^\delta = f$ on $\partial B_{1+\delta}$. Let $h_D^{n,\delta}$ denote the unique solution of

$$\begin{cases} \Delta^{\mathcal{G}_n} h_D^{n,\delta} = 0 & \text{in } \mathcal{V}_n[\mathbb{B}]\,, \\ h_D^{n,\delta} = h_C^\delta & \text{on } \partial\mathcal{V}_n[\mathbb{B}]\,. \end{cases}$$

By (4.2), for each $\delta > 0$,

$$\lim_{n\to\infty} \sup_{x\in\mathbb{B}} |h_D^{n,\delta}(x^n) - h_C^\delta(x)| = 0\,.$$

Also, by smoothness of f and the continuous dependence of harmonic functions on their boundary data, the maximum principle, and (4.1),

$$\lim_{n\to\infty} \lim_{\delta\to 0} \left(\left| h_D^{n,\delta}(x^n) - \int_{\mathbb{R}^d} f(z) d\,\mathrm{hm}^{\mathrm{n}}(x)[z] \right| + \left| \int_{\partial\mathbb{B}} f(z) d\,\mathrm{hm}(x)[z] - h_C^\delta(x) \right| \right) = 0\,.$$

Combining the previous two displays with the triangle inequality verifies (4.3), completing the proof. □

Since Assumption D is preserved under scaling and translation this also shows convergence of the harmonic measure of rescaled and translated balls.

4.2. **Convergence of random walk.** The argument given here is similar to the proofs of [Gwynne et al., 2022, Lemma 3.14 and Theorem 3.10].

Proof of Theorem 4.1. The idea is to apply rescaled and translated versions of Proposition 4.2 to a sequence of small balls following the path of the random walk.

Fix $z \in U$, a small parameter $\delta > 0$, and cover the domain $B_{2\delta}(U)$ by balls of radius δ centered at *grid points* defined as elements of the set $\delta\mathbb{Z}^d \cap B_{2\delta}(U)$.

We iteratively define, for $k \in \mathbb{N}$, a sequence of stopping times $J_k^{z,n}$ and grid points $w_{\delta,k}^{z,n}$. Start with $J_0^{z,n} = 0$ and choose $w_{\delta,0}^{z,n}$ so that the point $z^n \in \mathcal{V}_n[B_\delta(w_{\delta,0}^{z,n})]$. Then, having defined $w_{\delta,k-1}^{z,n}$ and $J_{k-1}^{z,n}$, if $w_{\delta,k-1}^{z,n} \notin B_\delta(U)$, then stop, setting $K_\delta^{z,n} := k-1$. Otherwise, let $w_{\delta,k}^{z,n}$ be the nearest (ties broken lexicographically) grid point to $Y_{J_{k-1}^{z,n}}^{z,n}$. Let $J_k^{z,n}$ be the smallest $t \geq J_{k-1}^{z,n}$ for which $X_t^{z,n} \in \partial\mathcal{V}_n[B_{2\delta}(w_{\delta,k}^{z,n})]$.

The corresponding Brownian motion stopping times are defined as follows. Initialize $\tau_{\delta,0}^z = 0$ and the grid point $w_{\delta,0}^z = w_{\delta,0}^{z,n}$. Then, having defined $w_{\delta,k-1}^z$ and τ_{k-1}^z, if $w_{\delta,k-1}^z \notin B_\delta(U)$, then stop, setting $K_\delta^z := k-1$. Otherwise, let $w_{\delta,k}^z$ be the nearest (ties broken lexicographically) grid point to $\mathcal{B}_{\tau_{k-1}^z}^z$. Let τ_k^z be the smallest $t \geq \tau_{k-1}^z$ for which $\mathcal{B}_t^z \in \partial B_{2\delta}(w_{\delta,k}^z)$.

Let $P_{\delta,k}^{z,n}$ denote the conditional law of $Y_{J_{\delta,k}^{z,n}}^{z,n}$ given $w_{\delta,k}^{z,n}$ and let $\mathrm{hm}_{\delta,k}^z$ be the (continuum) harmonic measure on $\partial B_{2\delta}(w_{\delta,k}^{z,n})$ as viewed from $Y_{J_{\delta,k-1}^{z,n}}^{z,n}$. By (rescaled and translated versions of) Proposition 4.2 and the strong Markov property of random walk,

$$\lim_{n\to\infty} \sup_{z\in U} \sup_{k\in[0,K_\delta^{z,n}]\cap\mathbb{N}} \mathrm{d}^{\mathrm{Pr}}\left(P_{\delta,k}^{z,n}, \mathrm{hm}_{\delta,k}^z\right) = 0\,.$$

From this, continuity of harmonic measure, and the fact that K^z_δ and $K^{z,n}_\delta$ are almost surely finite, we find that the supremum over all $z \in U$ of the Prokhorov distance between the laws of $\left(Y^{z,n}_{J^{z,n}_{\delta,k}}\right)_{k\in[0,K^{z,n}_\delta]\cap\mathbb{N}}$ and $\left(\mathcal{B}^z_{\tau^z_{\delta,k}}\right)_{k\in[0,K^z_\delta]\cap\mathbb{N}}$ tends to 0 as $n \to \infty$.

Also, by construction and (4.1) we have that

$$\sup_{s,t\in[\tau^z_{\delta,k-1},\tau^z_{\delta,k}]} \|\mathcal{B}^z_s - \mathcal{B}^z_t\| \le 4\delta \quad \text{and} \quad \lim_{n\to\infty} \sup_{s,t\in[J_{\delta,k-1},J_{\delta,k}]} \|Y^{z,n}_s - Y^{z,n}_t\ | \le 4\delta\,. \tag{4.4}$$

From this and the preceding paragraph, it follows that we can couple the laws of $Y^{z,n}$ and $\mathcal{B}^z$ in such a way that with probability at least $1-\delta$, we have $K^{z,n}_\delta = K^z_\delta$ and

$$\sup_{k\in[0,K^z_\delta]\cap\mathbb{N}} \left|Y^{z,n}_{J^{z,n}_{\delta,k}} - \mathcal{B}^z_{\tau^z_{\delta,k}}\right| \le \delta\,.$$

If this is the case, we re-parameterize $Y^{z,n}|_{[0,J^{z,n}_U]}$ in such a way that $[J^{z,n}_{\delta,k-1}, J^{z,n}_{\delta,k}]$ is traced in $\tau^z_{\delta,k} - \tau^z_{\delta,k-1}$ units of time. Since U is Lipschitz, a standard Brownian motion estimate shows that $|\tau_{\delta,K^z_\delta} - \tau^U_z| \le C\delta$ for some constant $C(U)$ with probability going to one as $\delta \to 0$. Consequently, with probability going to one as $\delta \to 0$, by (4.4), the uniform distance between this re-parameterized version of the curve $Y^{z,n}|_{[0,J^{z,n}_U]}$ and $B^z|_{[0,\tau^z_U]}$ is at most $C\delta$. Since δ and z were arbitrary, this completes the proof. □

5. Verifying the hypothesis

In this section we show that Voronoi tesselations of (possibly very degenerate) point processes satisfy Hypothesis (II). That is, we prove Proposition 1.1.

First, we show a deterministic property of Voronoi tesselations: if the points are sufficiently spread out, then the minimal cell size cannot be large, and if the points are not concentrated, then the maximal cell size cannot be small. In the statement and below, we denote by $[-r,r]^d$ the d-dimensional cube of radius $r > 0$ centered at the origin.

Lemma 5.1. *Let $U \subset \mathbb{R}^d$ be a bounded domain and let $S \subset \mathbb{R}^d$ be a locally finite set of points*

(a) If there exists $r > 0$ with at most one point in $x + [-2r, 2r]^d$ for all $x \in r\mathbb{Z}^d \cap U$, then

$$B_r(s) \subset \mathcal{C}_s \quad \forall s \in S \cap U\,.$$

(b) There exists a constant $\delta = \delta(d) \in (0,1)$ such that if there exists $k > 0$ with at least one point in $y + [-\delta k, \delta k]^d$ for all $y \in k\mathbb{Z}^d \cap B_{2k}(U)$, then

$$\mathcal{C}_s \subset B_{4k\sqrt{d}}(s) \quad \forall s \in S \cap U\,.$$

To prove Lemma 5.1, we use a particular case of the following fact: if a convex polytope is large, then the dual polytope must be small.

Lemma 5.2. *There exists a constant $\delta'(d) \in (0,1)$ such that the following holds for every selection of $x^{(i^\pm)} \in [-\delta', \delta']^d \pm e_i$ for $i = 1, \dots, d$. The dual of the convex polytope $P := \mathbf{conv}(x^{(1^+)}, x^{(1^-)}, \dots, x^{(d^+)}, x^{(d^-)})$ defined by $P^* := \{y \in \mathbb{R}^d : x \cdot y \le 1 \text{ for all } x \in P\}$ is bounded by $P^* \subset B_{2\sqrt{d}}$.*

Proof. It suffices to show that the ball of radius $R := (4d)^{-1/2}$ centered at the origin is contained in the polytope, $B_R \subset P$. Indeed, by the definition of dual, we have that $P^* \subset B^*_R$ and since $B^*_R = B_{R^{-1}}$, this would show $P^* \subset B_{2\sqrt{d}}$.

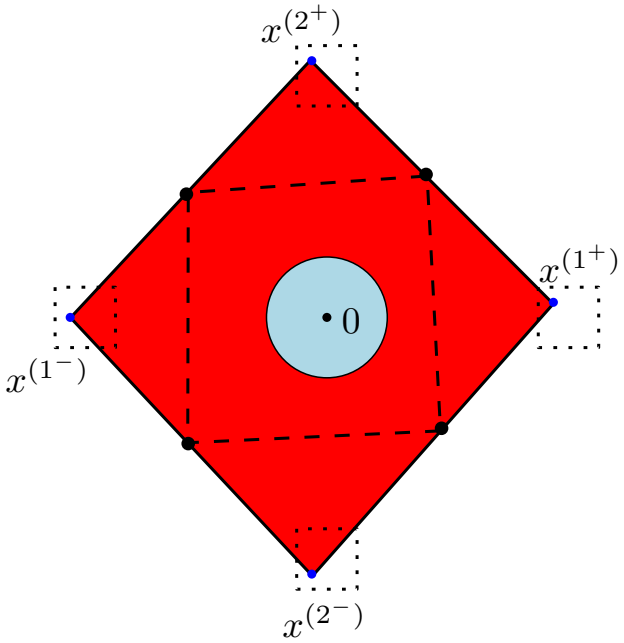


FIGURE 12. Illustration of the proof of Lemma 5.2. The polytope P is in red and the dual P^* is outlined by a black dashed line. The cubes $[-\delta,\delta]^d \pm e_i$ containing $x^{(i^\pm)}$, which are blue dots, are outlined by a black dotted line.

To see that $B_R \subset P$, we first observe, by the Cauchy-Schwartz inequality, that the ball of radius $d^{-1/2}$ is contained in the convex hull of $\mathbf{conv}(\pm e_1, \dots, \pm e_d)$; indeed, if $z \in B_1$,

$$\left(\sum_{i=1}^d |z_i|\right)^2 \leq d \sum_{i=1}^d z_i^2 \leq d\,.$$

Since the map from the set of points to their convex hull is Lipschitz continuous with respect to the Hausdorff distance, this shows that for small enough δ', depending only on d, the ball of radius $(4d)^{-1/2}$ is contained in P. This completes the proof. □

Proof of Lemma 5.1. Proof of (a). Let $s \in S \cap U$. Let s' be the nearest, in Euclidean distance, point in S to s. Let $x \in r\mathbb{Z}^d \cap U$ be such that $\|s - x\| \leq r$. Then, since there can be no other points in $x + [-2r, 2r]^d$, we must have $\|s - s'\| > r$.

Proof of (b). Let $s \in S \cap U$ and let $\delta := c\delta'$ where δ' is as in Lemma 5.2 and $c(d) \in (0,1)$ is a constant to be determined below. By rescaling and translation, we may suppose $s = 0$ and $k = \frac{1}{2}$, and that there are, for each $i \in \{1, \dots, d\}$, points $s^{(i^\pm)} \in \{[-cd\delta', cd\delta']^d \pm \frac{1}{2}e_i\} \cap S$. Denote this set of points by S'. Since $s = 0$, we can rewrite, for all $s' \in S'$ and $y \in \mathbb{R}^d$, the halfspace

$$(y - \frac{1}{2}(s + s')) \cdot (s - s') \geq 0 \iff -y \cdot s' + \frac{1}{2}\|s'\|^2 \geq 0 \iff y \cdot \left(2\frac{s'}{\|s'\|^2}\right) \leq 1\,,$$

so that by (1.6) and the fact $S' \subset S$, we have

$$\mathcal{C}_s \subset \bigcap_{s' \in S'} \left\{ y \in \mathbb{R}^d : y \cdot \left(2\frac{s'}{\|s'\|^2}\right) \leq 1 \right\}\,.$$

Define $x^{(i^\pm)} := 2\frac{s^{(i^\pm)}}{\|s^{(i^\pm)}\|^2}$ and observe that for a sufficiently small choice of c, we have $x^{(i^\pm)} \in [-\delta', \delta']^d \pm e_i$ for each $i \in \{1, \dots, d\}$. Thus, Lemma 5.2 yields that $\mathcal{C}_s \subset B_{2\sqrt{d}}$. This completes the proof after undoing the scaling. □

It remains to show that the conditions of Lemma 5.1 are satisfied with probability one as $m \to \infty$ for particular choices of the grid.

Proof of Proposition 1.1. We will show that the events in Lemma 5.1 are satisfied for

$$k := m^{-1/(2\beta^+)} \quad \text{and} \quad r := m^{-8/\beta^-}\,, \tag{5.1}$$

for all large enough m for a slightly enlarged set $U' := B_{4m^{-1/(4\beta^+)}}(U)$. For a finite set of points we write $|\cdot|$ for the cardinality. We use the fact that the number of points in a cube of side length s, the random variable $|\Lambda_m \cap [-s, s]^d|$, is Poisson with mean $m\mu([-s, s]^d)$.

For the lower bound, let $\delta(d)$ be as in Lemma 5.1 so that by (1.8), for all m sufficiently large,

$$m\mu([-\delta k, \delta k]^d) \geq m^{1/3}\,,$$

and so

$$\mathbb{P}[|\Lambda_m \cap [-\delta k, \delta k]| = 0] = \exp(-m\mu([-\delta k, \delta k]^d)) \leq \exp(-m^{1/3})$$

which shows, after taking a union bound,

$$\mathbb{P}\left[\bigcup_{y \in k\mathbb{Z}^d \cap B_{2k}(U')} \{\text{the cube } \{y + [-\delta k, \delta k]^d\} \text{ contains no points of } \Lambda_m\}\right] \leq \exp(-m^{1/4})\,. \tag{5.2}$$

For the upper bound, we first observe that if $X \sim \text{Poisson}(\lambda)$, then, for all λ sufficiently small,

$$\mathbb{P}[X \geq 2] = 1 - \mathbb{P}[X = 0] - \mathbb{P}[X = 1] = 1 - \exp(-\lambda) - \lambda\exp(-\lambda) \leq \lambda^2\,. \tag{5.3}$$

Fix $N \geq 1$ to be selected below, and let

$$\beta^- = b_0 < \cdots < b_N = \beta^+$$

be a partition of the interval $[\beta^-, \beta^+]$. For each $j \in \{1, \dots, N\}$, let R_j be the set of grid boxes of radius $2r$ with μ-volume between $(2r)^{b_j}$ and $(2r)^{b_{j-1}}$,

$$R_j := \{B := y + [-2r, 2r]^d : y \in r\,\mathbb{Z}^d \cap U' \text{ and } \mu(B) \in [(2r)^{b_j}, (2r)^{b_{j-1}}]\}\,.$$

Observe that by (1.8), for every $j \in \{1, \dots, N\}$ we have

$$|R_j| \leq r^{-b_j} d^d \mu(U')\,, \tag{5.4}$$

and also,

$$|\Lambda_m \cap B| \sim \text{Poisson}(\lambda_j) \quad \forall B \in R_j \quad \text{with } \lambda_j \leq m2^{\beta^+} r^{b_{j-1}}\,.$$

By the previous display and (5.3) we have, for all $j \in \{1, \dots, N\}$,

$$\mathbb{P}[|\Lambda_m \cap B| \geq 2] \leq m^2 4^{\beta^+} r^{2b_{j-1}} \qquad \forall B \in R_j$$

and so by a union bound and (5.4),

$$\mathbb{P}\left[\bigcup_{B \in R_j} \{\text{the cube } B \text{ contains at least two points of } \Lambda_m\}\right] \leq m^2 4^{\beta^+} r^{2b_{j-1} - b_j} \mu(U')\,.$$

By taking N sufficiently large, depending only on β^+ and β^- so that $b_j - b_{j-1} \le \beta^-/2$, we have $m^2 r^{2b_{j-1}-b_j} \le m^2 r^{\beta^-/2}$. Consequently, by the previous display and a union bound over j, we have

$$\mathbb{P}\left[\bigcup_{y \in r\mathbb{Z}^d \cap U'} \left\{\text{the cube } \{y + [-2r, 2r]^d\} \text{ contains at least two points of } \Lambda_m\right\}\right] \le N d^d 4^{\beta^+} m^2 r^{\beta^-/2} \mu(U') \le N d^d 4^{\beta^+} m^{-2} \mu(U'),$$

and so for m sufficiently large

(5.5)
$$\mathbb{P}\left[\bigcup_{y \in r\mathbb{Z}^d \cap U'} \left\{\text{the cube } \{y + [-2r, 2r]^d\} \text{ contains at least two points of } \Lambda_m\right\}\right] \le m^{-3/2}.$$

Combining (5.5) and (5.2) together with the Borel-Cantelli lemma shows that the hypotheses required for Lemma 5.1 are satisfied almost surely for all m sufficiently large for U'. Since every such cell which intersects U must be centered at a grid point in U', this completes the proof of (1.9).

This also shows that $\varepsilon_m := \sup_{v \in \mathcal{V}^{\mathrm{Vor}}(\Lambda_m)} \operatorname{diam}(P_v) \le m^{-1/(3\beta^+)}$ for all large m and that (1.3) is satisfied almost surely. To see that Hypothesis (II) is satisfied, we observe that (1.9) implies that the minimum volume of a cell in $\mathcal{G}_m$ is $m^{-d8/\beta^-} \ge \varepsilon_m^{d(3\beta^+)8/\beta^-}$. □

6. Counterexample for simple random walk

It is straightforward to construct, in all dimensions, sphere packings of *unbounded* degree upon which simple random walk with unit conductance does not converge to Brownian motion modulo time change, see, for example, [Nachmias, 2020, Example 4.8]. In this section we give a *bounded* degree example in dimensions at least three. This is similar to the potential examples discussed in [Gwynne et al., 2022, Section 1.5]. We are not sure whether such a counterexample exists in two dimensions.

Theorem 6.1. *For all $d \ge 3$, there exists a sphere packing of $\mathbb{R}^d$ upon which simple random walk with unit conductance does not converge to Brownian motion modulo time parametrization in the scaling limit. Moreover, the spheres all have radii either 1 or 1/2 and are tangent to a bounded number of other spheres.*

The sphere packing we construct will force the random walk to lack negation symmetry at large scales, precluding it from converging to Brownian motion.

Proof of Theorem 6.1. We start with the construction of the sphere packing, part of which is illustrated in Figure 13.

- At each vertex x on $2\mathbb{Z}^d$ for $x_1 \le 0$, place a *big* sphere of radius 1.
- The big spheres centered at $x \in \mathbb{Z}^d$ for $x_1 = 0$ are the big *middle* spheres.
- Place *small* spheres of radius $\frac{1}{2}$ at $(\frac{\sqrt{7}}{2}, \frac{1}{2}, \frac{1}{2}, 0, \dots, 0) + y$ for $y \in \mathbb{Z}^d$ with $y_1 \ge 0$.
- The small spheres with centers x for $x_1 = \frac{\sqrt{7}}{2}$ are *small middle* spheres.

Two spheres are adjacent to each other if and only if the distance between their centers is equal to the sum of the two radii. Thus, non-middle spheres are adjacent to $2d$ other

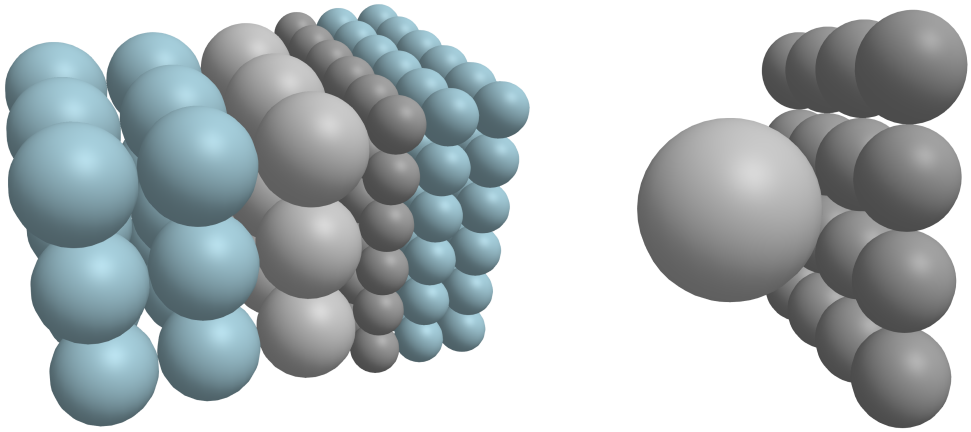


FIGURE 13. On the left, part of the sphere packing given in Theorem 6.1 for $d = 3$. On the right, a big middle sphere and some nearby middle small spheres. Big (resp. small) middles spheres are in light (resp. dark) gray and the other spheres are in light blue.

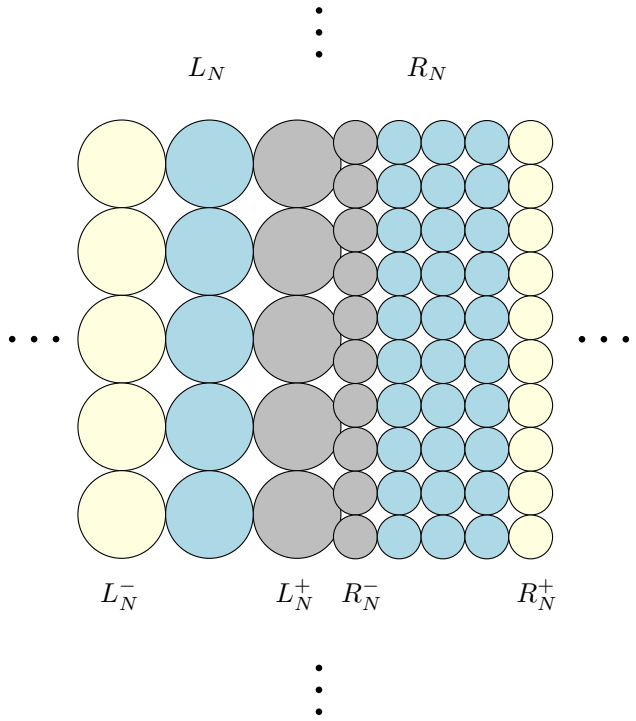


FIGURE 14. A top-down view of the sphere packing given in Theorem 6.1 for $d = 3$. The spheres in L_N, R_N (defined in (6.1)) are in light blue, the spheres in L_N^-, R_N^+ are in light yellow, the spheres in L_N^+, R_N^- are in gray (defined in (6.2)). Here $N = 2$. Note that the small middle spheres are slightly above small big spheres so they are not overlapping.

spheres of the same size. Also, by construction, each big middle sphere is adjacent to $2d-1$ big spheres and 4 small middle spheres and each small middle sphere is adjacent to $2d-1$ small spheres and 1 big middle sphere, see Figure 13. Indeed, the big middle sphere with center $(0, x_2, \dots, x_d)$ is adjacent to the four small middle spheres with centers $(\frac{\sqrt{7}}{2}, x_2 \pm \frac{1}{2}, x_3 \pm \frac{1}{2}, x_4, \dots, x_d)$ as

$$\sqrt{\left(\frac{\sqrt{7}}{2}\right)^2 + \left(\pm\frac{1}{2}\right)^2 + \left(\pm\frac{1}{2}\right)^2} = 3/2 = 1 + 1/2\,.$$

Denote the sphere packing by $\mathcal{S}$ and identify spheres in $\mathcal{S}$ by their centers.

We now argue that simple random walk (SRW) on the tangency graph of $\mathcal{S}$ is more likely to go right than to go left when started in the middle. To that end, we consider the x_1 coordinate of the walk. Fix $N \geq 2$ and divide space according to Figure 14 as follows:

$$L_N = \{x \in \mathcal{S} : -2N < x_1 < 0\} \quad \text{and} \quad R_N = \{x \in \mathcal{S} : \frac{\sqrt{7}}{2} < x_1 < 2N + \frac{\sqrt{7}}{2}\}\,, \tag{6.1}$$

and consider their boundaries

$$\begin{aligned} L_N^- &= \{x \in \mathcal{S} : x_1 = -2N\} \qquad L_N^+ = \{x \in \mathcal{S} : x_1 = 0\} \\ R_N^- &= \{x \in \mathcal{S} : x_1 = \frac{\sqrt{7}}{2}\} \qquad R_N^+ = \{x \in \mathcal{S} : x_1 = 2N + \frac{\sqrt{7}}{2}\}\,. \end{aligned} \tag{6.2}$$

A standard estimate for simple random walk on $\mathbb{Z}^d$ (obtained by considering the harmonic function $x \mapsto x_1$) gives

$$\mathbb{P}\left[\text{SRW started at } \{x_1 = -2\} \text{ hits } L_N^- \text{ before } L_N^+\right] = \frac{1}{N}$$

and

$$\mathbb{P}\left[\text{SRW started at } R_N^- \text{ hits } L_N^+ \text{ before } R_N^+\right] = \frac{2N}{2N+1}\,.$$

Putting these estimates together and denoting

$$p := \mathbb{P}[\text{SRW started at } L_N^+ \text{ hits } L_N^- \text{ before } R_N^+]$$

we have, by conditioning on the first exit from L_N^+, using the fact that each big middle sphere is adjacent to 4 small middle spheres,

$$\begin{aligned} p = {} & \mathbb{P}\left[\text{SRW started at } \{x_1 = -2\} \text{ hits } L_N^- \text{ before } L_N^+\right] \frac{1}{5} \\ & + p\mathbb{P}\left[\text{SRW started at } \{x_1 = -2\} \text{ hits } L_N^+ \text{ before } L_N^-\right] \frac{1}{5} \\ & + p\mathbb{P}\left[\text{SRW started at } R_N^- \text{ hits } L_N^+ \text{ before } R_N^+\right] \frac{4}{5} \\ = {} & \frac{1}{5}\left(\frac{1}{N} + p\frac{N-1}{N}\right) + \frac{4}{5}p\frac{2N}{2N+1}\,. \end{aligned}$$

Solving the above equation yields that

$$p = \frac{2N+1}{6N+1}\,,$$

in particular,

$$\lim_{N\to\infty} \mathbb{P}[\text{SRW started at } (0,\dots,0) \text{ hits } R_N^+ \text{ before } L_N^-] = \frac{2}{3}\,.$$

Since Brownian motion satisfies negation symmetry, this is incompatible with the SRW converging to Brownian motion, modulo time parameterization. □

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